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Thermodynamics of Fe–Sm, Fe–H, and H–Sm systems and its application to the hydrogen–disproportionation–desorption–recombination (HDDR) process for the system $Fe_{17}Sm_2$–H_2

M. Zinkevich*, N. Mattern, A. Handstein, O. Gutfleisch

Institut für Festkörper- und Werkstofforschung, Postfach 270016, 01171 Dresden, Germany

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Abstract

Thermodynamic properties of the Fe–Sm, Fe–H, and H–Sm systems have been analyzed by means of the CALPHAD method. Thermodynamic models have been defined to describe the Gibbs energy of the individual phases and the model parameters have been optimized using different experimental information: phase diagram data, calorimetric data, and equilibrium partial pressures of hydrogen. The entropy of $Fe_{17}Sm_2$ has been derived from experimental data on low-temperature heat capacity. The thermodynamic parameters for $Fe_{17}Sm_2$ have been evaluated using binary (Fe–Sm) and ternary ($Fe_{17}Sm_2$–H_2) data simultaneously. The calculated phase diagrams and thermodynamic properties are in agreement with experiments. The thermodynamics of hydrogen in the interstitial solid solution $Fe_{17}Sm_2H_\delta$ have been described by compound energy formalism. Combining the obtained thermodynamic descriptions of the individual phases, the equilibrium temperatures of the recombination reaction: $Fe+SmH_{2\pm x}\Rightarrow Fe_{17}Sm_2+H_2$, which is an important part of the HDDR process, have been calculated as a function of hydrogen pressure. © 2002 Elsevier Science B.V. All rights reserved.

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1. Introduction

Interest in the thermodynamics of hydrogen in metals and alloys has been increasing since there are many practical uses of metal–hydrogen systems such as storage of hydrogen for clean energy purposes or absorption of nuclear radiation [1]. The intermetallic compounds based on rare earth and transition metals have attracted special interest not only due to their hydrogen storage properties, but also due to the significant changes in intrinsic magnetic properties upon dissolution of hydrogen. For example, hydrogen in $Fe_{17}Sm_2$ increases the Curie temperature by about 140 K [2].

Furthermore, the reaction with hydrogen can be used to obtain a fine-grained microstructure by means of the hydrogenation–disproportionation–desorption–recombination (HDDR) process and thereby improving the extrinsic magnetic properties. For instance, the heat treatment of $Fe_{17}Sm_2$ compound in a hydrogen atmosphere results in formation of SmH_2 grains of several nanometers in diameter within the α-Fe matrix. After recombination in vacuum, the grain-refined powder consisting of the reformed parent phase gives high values of coercivity upon a nitrogenation process to $Fe_{17}Sm_2N_3$ [3–7].

The knowledge of the phase equilibria is essential for the optimization of magnetic properties and the CALPHAD method [8,9] (an acronym for calculation of phase diagrams by computer coupling of phase diagrams and thermochemistry) is widely used in materials science and engineering. This technique allows the calculation of any kind of phase diagram and thermodynamic properties of interest based on self-consistent thermodynamic description of the system. In order to obtain a thermodynamic description of a multicomponent system, a thermodynamic description of each lower-order system is necessary. The phase diagram of the Fe–Sm system was calculated previously in the work of Schneider [10] but the system had to be reassessed since old unary data were used [11] and neither polymorphism of samarium nor magnetic contributions were taken into account. In this assessment

*Corresponding author. Present address: Max-Planck Institute for Metal Research, Heisenbergstr. 5, 70569 Stuttgart, Germany.

E-mail address: zinkevich@mf.mpg.de (M. Zinkevich).

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only constitutional data were used. It is, however, always desirable to use also thermochemical data in the optimization to make sure that coefficients are correctly determined. Since no measurements of thermodynamic properties were reported in the literature, the low-temperature heat capacity of $Fe_{17}Sm_2$ was measured in the present work. For the Fe–H and H–Sm systems, no thermodynamic assessments were found in the literature and they had to be obtained for the first time in this study.

The aim of the present work is to obtain self-consistent thermodynamic descriptions of the binary systems Fe–Sm, Fe–H, and H–Sm, while the available literature data should be critically assessed and checked for their mutual consistency, and then, to calculate the phase diagram for the system $Fe_{17}Sm_2$–H_2.

2. Experimental

The heat capacity (C_P) of $Fe_{17}Sm_2$ has been measured from 5 to 298 K using the PPMS Quantum Design instrument. The data in the region $T \leq 20$ K were fit to $C_P = \gamma T + 1944.6T^3/\Theta_D^3$, where γ is an electronic specific heat constant and Θ_D is the Debye temperature at $T=0$ K. The standard entropy $^\circ S_{298}$ ($Fe_{17}Sm_2$) was obtained by the numerical integration under the curve C_P/T vs. T from 0 to 298.15 K.

3. Assessment of the literature information

3.1. The Fe–Sm system

The only experimental study of the Fe–Sm phase diagram is the work of Buschow [12], who used X-ray diffraction, thermoanalysis (DTA) and metallography to determine the crystal structure of phases, the kind and temperatures of invariant reactions as well as several points on the liquidus curve. There are three stoichiometric intermetallic compounds in this system: $Fe_{17}Sm_2$ (Th_2Zn_{17}-type), Fe_3Sm ($PuNi_3$-type), and Fe_2Sm ($MgCu_2$-type), formed by peritectic reactions. All compounds show a ferromagnetic behaviour [12–42].

Samata and co-workers [43,44] prepared single crystals of Fe_7Sm and $Fe_{23}Sm_6$ by means of a modified self-flux method. However, it remains still unclear, whether these compounds are thermodynamically stable, since no thermal analysis has been performed. Therefore, the existence of Fe_7Sm and $Fe_{23}Sm_6$ was not taken into account in the present work. Liu and Jin [45] studied phase equilibria in the Sm–Fe–Ti system by using diffusion triple technique and electron microprobe analysis (EPMA). After 500 h annealing at 600 °C only three intermetallic phases $Fe_{17}Sm_2$ (10.0–10.9 at% Sm), Fe_3Sm (24.0–26.8 at% Sm), and Fe_2Sm (32.7–33.8 at% Sm) were found between Fe and Sm pieces. Contrary to Ref. [12], the measurements of Ref. [45] showed an appreciable homogeneity range for each compound (the numbers in parentheses). The maximum solubility of Sm in α-Fe is very small (about 0.3 at%) and the solubility limit of Fe in α-Sm was even below the detection limit of the EPMA.

3.2. The H–Sm system

The H–Sm system was recently reviewed by Manchester and Pitre [46]. The assessed temperature–composition (T–x) phase diagram for the pressures around 100 kPa includes the following phases:

- (α-Sm), a random interstitial alloy of H in the complex rhombohedral structure of α-Sm (space group $R\bar{3}m$), existing from pure samarium to a maximum concentration of about 8 at% H.
- (β-Sm) and (γ-Sm), random interstitial alloys of H in the hexagonal close packed (h.c.p.) structure of β-Sm and the body centered cubic (b.c.c.) structure of γ-Sm, respectively, both extending to concentrations above 30 at% H. The β⇔γ transformation temperature of pure Sm is almost unchanged upon alloying with hydrogen.
- $SmH_{2\pm x}$ phase, a non-stoichiometric samarium dihydride with the face centered cubic (f.c.c.) fluorite-type structure.
- SmH_{3-y} phase, a non-stoichiometric samarium trihydride. Two types of hexagonal structure were reported for this phase: HoH_3-type, space group $P3c1$ and LaF_3-type, space group $P6_3/mmc$.
- Liquid phase, dissolving more than 30 at% H. The melting temperature of Sm is not lowered with the addition of hydrogen.
- Gas phase, which consists of the molecular hydrogen H_2 and atomic species H and Sm.

The pressure–composition (pH_2–x) phase diagram for the H–Sm system shows two plateaus in the (β-Sm)+SmH_{2-x} and SmH_{2+x}+SmH_{3-y} mixed phase regions. Single $SmH_{2\pm x}$ phase is stable between $2-x \approx 1.8$ up to $2+x \approx 2.5$. When $x<0$ only the tetrahedral sites in the f.c.c. lattice are occupied by H atoms, while for $x>0$ additional hydrogens occupy the octahedral voids. The homogeneity region of SmH_{3-y} extends from $y \approx 0.1$ up to the highest attainable H concentrations ($y=0$).

There have been no calorimetric determinations of the thermodynamic quantities, except the enthalpy of reaction: $SmH_{2\pm x}+H_2 \Rightarrow SmH_{3-y}$ measured by DTA [47,48]. So far, most of the calorimetric measurements on the H–Sm system were concerned with low-temperature specific heat of SmH_2 and $SmH_{2.84}$ ($T<240$ K). For the former phase the anomaly around $T_N=7$ K due to the antiferromagnetic ordering was detected [46].

3.3. The Fe–H system

The (T–x) phase diagram for the Fe–H system is rather

simple. The hydrogen solubility in iron is very limited (<0.2 at% at 100 kPa H_2), and the only observed iron hydride FeH forms at the hydrogen pressure of more than 6 GPa [49], which was beyond the scope of the present study. The experimental data for the hydrogen solubility in iron at the pressure of 101 325 Pa have been compiled [50,51]. Almost all the measurements from many literature sources agree well with each other in the region $T>$ 500 °C, whereas the degree of scatter increases rapidly as the temperature decreases. By close examination of the experimental technique used in different works and analyzing of possible sources of errors, the 'best' solubility data for b.c.c. Fe have been selected by Ref. [50]. The general trend for the limited solubility to grow with increasing temperature is influenced by the allotropic transformations in solid iron, while the solubility becomes almost twice as large above the melting point. The abrupt increase of hydrogen solubility due to the $\alpha \Rightarrow \gamma$ transition in Fe was unambiguously confirmed by many investigations. At the same time, not all authors who studied the solubility up to the melting point of iron could find a small jump, which is related to the $\gamma \Rightarrow \delta$ transition. The solubility becomes indeed smaller in δ-Fe, but this phenomenon is difficult to observe, since the hydrogen content in the solid iron increases more and more rapidly with temperature. It has been pointed out [50,52] that the plot of logarithm of the hydrogen solubility versus reciprocal temperature is non-linear. Da Silva and McLellan [52] suggested two possible explanations of such departure from Arrhenius behaviour. On the one hand, the enthalpy and entropy of solution of hydrogen in iron may be temperature-dependent. On the other hand, hydrogen may occupy both the tetrahedral and octahedral interstitials in the b.c.c. iron lattice simultaneously.

Hill and Johnson [53] measured the hydrogen solubility in iron at pH_2 up to 13.8 MPa and found the approximate conformance of the solubility to Sieverts law, i.e. a proportionality between the concentration of dissolved H and the square root of hydrogen pressure. A study of hydrogen solubility in iron at higher pressures (up to 100 MPa) showed negative deviations from Sieverts law at $T>700$ °C and $P>10$ MPa [54]. In addition, the temperatures of the allotropic transformations of iron were measured as functions of hydrogen pressure [54–56]. The dissolved hydrogen lowers the melting point of iron and extends the temperature range, where the f.c.c. Fe exists. The solubility and the critical temperatures approach saturation above 20–30 MPa H_2. A 'limiting' phase diagram Fe–H has been suggested for the hydrogen pressure of 40 MPa [54]. The maximum solubility (in the liquid iron) approaches 2 at% H.

3.4. The $Fe_{17}Sm_2$–H_2 system

Koeninger et al. [57] studied the hydrogen sorption characteristics of $Fe_{17}Sm_2$ using a Sieverts-type apparatus with powder samples at temperatures between 299 and 773 K and pressures between 1 kPa and 7 MPa. No sign of a distinct hydride phase within hydrogen concentrations $\delta<$ 4.5 in $Fe_{17}Sm_2H_\delta$ was found. They also detected the instability of $Fe_{17}Sm_2H_\delta$ above 673 K after 12 h hydriding–dehydriding treatment and suggested that it can be unstable under the experimental conditions.

Christodoulou and Takeshita [19] studied the reaction between $Fe_{17}Sm_2$ and H_2 in the temperature range from 298 to 1273 K by means of isochorothermal analysis, X-ray diffraction and thermomagnetic analysis using different initial H_2 pressures (0–125 kPa). According to their results, absorption and desorption of hydrogen by $Fe_{17}Sm_2H_\delta$ is completely reversible between 298 and 773 K. At higher temperatures the following disproportionation reaction occurs: $Fe_{17}Sm_2H_\delta + (2+x-\delta/2)H_2 \Rightarrow 2SmH_{2+x} + 17Fe$ ($\delta \approx 0.1$). A more recent study of the hydrogen absorption behavior of $Fe_{17}Sm_2$ at hydrogen pressures between 50 kPa and 0.8 MPa by means of differential thermal analysis showed that increased hydrogen pressures promote the disproportionation reaction, so that the temperature, at which the exothermic effect occurs, decreases [58]. The maximum interstitial hydrogen concentration in $Fe_{17}Sm_2H_\delta$ was found to be $\delta=5$ [19,59]. The Curie temperature and the saturation magnetization of fully hydrogenated compound $Fe_{17}Sm_2H_5$ have been measured [19] (see below). Teresiak et al. [60] determined the hydrogen content in $Fe_{17}Sm_2H_\delta$ after 3 h isothermal annealing at 453 K and $pH_2=40$–80 kPa using three different methods, which resulted in the mean value $\delta=$ 4.0±0.8.

Above 923 K, the disproportionated material desorbs H_2 and recombines to form $Fe_{17}Sm_2$ [19,30,61–63]. The equilibrium temperature of the invariant recombination reaction: $Fe + SmH_{2\pm x} \Rightarrow Fe_{17}Sm_2 + H_2$ depends on the hydrogen partial pressure in the gas phase increasing from 923 K for vacuum conditions [30] up to 1313 K for 101 325 Pa H_2 [61]. Despite the large number of publications, in which the HDDR of $Fe_{17}Sm_2$ has been studied, there are only two works where the recombination temperature was determined at conditions close to equilibrium [61,63].

4. Thermodynamic modeling

In order to maintain consistency with other thermodynamic assessments, the Gibbs energy of each element should be defined with respect to its stable state at 298.15 K and 101 325 Pa [64] as recommended by SGTE (Scientific Group Thermodata Europe). The Gibbs energy of stoichiometric phases and end-members of solutions is represented as a power series in terms of temperature in the form:

$$^\circ G(T) = a + bT + cT\ln(T) + dT^2 + eT^3 + f/T + \sum_n g_n T^n \quad (1)$$

where a to f and g_n are coefficients and n represents a set of integers. Pressure dependence of the Gibbs energy was only taken into account for the vapor phase (see below). For condensed phases, the variation of the Gibbs energy with pressure is negligible in the pressure range of interest ($P<10$ MPa). The reference states for the pure iron, samarium, and hydrogen are 1 mol of the b.c.c. Fe, rhombohedral Sm and $\frac{1}{2}$ mol of H_2 gas, respectively. The corresponding coefficients for condensed phases in unary systems were taken from the compilation of Dinsdale [64], except for β-Sm. The latter is not considered as a stable phase in Ref. [64], but it is commonly accepted now that the h.c.p. structure for pure samarium occurs from 1007 to 1195 K [65]. In the present work, the lattice stability for β-Sm was adopted from the recent assessment of Co–Sm system [66]. The Gibbs energy functions for the gaseous state are from JANAF tables [67].

The gas phase containing H_2, H, Fe, and Sm was treated as ideal mixture. Furthermore, an ideal gas behaviour was assumed in the pressure range of interest. The Gibbs energy of 1 mol of the gas phase is then expressed as:

$$G^{\text{gas}}_{(\text{H}_2,\text{H,Fe,Sm})} = \sum_i x_i G^{\text{gas}}_i + RT \sum_i x_i \ln x_i;$$
$$G^{\text{gas}}_i(P) = {}^{\circ}G^{\text{gas}}_i + RT \ln(P/P_0) \tag{2}$$

where ${}^{\circ}G^{\text{gas}}_i$ is the standard Gibbs energy of a gas, which contains only H_2, H, Fe, or Sm, respectively, and P_0 is the standard pressure of 101 325 Pa. R is the gas constant and x_i is the mol fraction of the corresponding specie.

The liquid phase was described by a two-sublattice model based on compound energy formalism [68] with one sublattice fully occupied by Fe and Sm and the other one where hydrogen and vacancies (Va) mix randomly, i.e. $(\text{Fe,Sm})_1(\text{H,Va})_1$. The number of sites in the second sublattice was arbitrarily fixed at the equiatomic composition, which allowed to reproduce all the experimental information. The Gibbs energy of the liquid phase is then expressed as follows:

$$\begin{aligned} G^{\phi}_{(\text{Fe,Sm}):(\text{H,Va})} &= y_{\text{Fe}} y_{\text{Va}} G^{\phi}_{\text{Fe:Va}} + y_{\text{Fe}} y_{\text{H}} G^{\phi}_{\text{Fe:H}} \\ &\quad + y_{\text{Sm}} y_{\text{Va}} G^{\phi}_{\text{Sm:Va}} + y_{\text{Sm}} y_{\text{H}} G^{\phi}_{\text{Sm:H}} + G^{\phi}_{\text{id}} \\ &\quad + G^{\phi}_{\text{ex}} \end{aligned}$$

$$G^{\phi}_{\text{id}} = RT(y_{\text{Fe}} \ln y_{\text{Fe}} + y_{\text{Sm}} \ln y_{\text{Sm}} + y_{\text{Va}} \ln y_{\text{Va}} + y_{\text{H}} \ln y_{\text{H}})$$

$$\begin{aligned} G^{\phi}_{\text{ex}} &= y_{\text{Fe}} y_{\text{Sm}} y_{\text{Va}} \sum_{\nu=0}^{n} L^{\phi}_{\text{Fe,Sm:Va}} (y_{\text{Fe}} - y_{\text{Sm}})^{\nu} \\ &\quad + y_{\text{Fe}} y_{\text{Sm}} y_{\text{H}} \sum_{\nu=0}^{n} L^{\phi}_{\text{Fe,Sm:H}} (y_{\text{Fe}} - y_{\text{Sm}})^{\nu} \\ &\quad + y_{\text{Fe}} y_{\text{Va}} y_{\text{H}} \sum_{\nu=0}^{n} L^{\phi}_{\text{Fe:H,Va}} (y_{\text{H}} - y_{\text{Va}})^{\nu} \\ &\quad + y_{\text{Sm}} y_{\text{Va}} y_{\text{H}} \sum_{\nu=0}^{n} L^{\phi}_{\text{Sm:H,Va}} (y_{\text{H}} - y_{\text{Va}})^{\nu} \end{aligned} \tag{3}$$

The colons separate elements on different sublattices. The term y_i represents the site fraction of component i on the second sublattice, while $G^{\phi}_{\text{Me:Va}}$ and $G^{\phi}_{\text{Me:H}}$ are the expressions for the Gibbs energy of the pure liquid elements and hypothetical associates MeH (Me stands for Fe or Sm). The Redlich–Kister parameters $L^{\phi}_{\text{Fe,Sm:Va}}$, $L^{\phi}_{\text{Fe,Sm:H}}$ and $L^{\phi}_{\text{Fe:H,Va}}$, $L^{\phi}_{\text{Sm:H,Va}}$ of an order ν represent the interaction energy between Fe and Sm atoms in the first sublattice and between hydrogen atoms and vacancies in the second sublattice, respectively.

It is also possible to adopt a substitutional solution model as proposed for Ni–H and Mg–H binary systems [69,70], in which Fe, Sm, and H are mixed together. Both models were tested, and the sublattice model was selected. This choice is supported by the fact that the experimental solubility range for the liquid phase in the H–Sm system could not be reproduced within the substitutional model using the assessed value for the Gibbs energy of the hypothetical liquid hydrogen G^{ϕ}_{H} [70]. Furthermore, the thermodynamic descriptions of the liquid phase in Fe–H and H–Sm systems could not be combined because they required different G^{ϕ}_{H} values. It seems to be the case due to increasing polarity of the chemical bonding between metal and hydrogen atoms in H–Sm system compared to that in the Fe–H, Ni–H, or Mg–H system. Note, that the maximum hydrogen content in the liquid phase of the Ni–H system does not exceed 0.5 at% at atmospheric pressure [70], while the corresponding value for the H–Sm system is about 100 times higher. The sublattice model is normally used for ionic liquids [68], but in the present work, it was extended to the metal–hydrogen systems, where no charge is taken into account. Using the sublattice model for the liquid phase allows to define individual $G^{\text{liq.}}_{\text{Me:H}}$ for each metal–hydrogen system instead of a single value $G^{\text{liq.}}_{\text{H}}$ for liquid hydrogen. Furthermore, any existing thermodynamic description in terms of a substitutional model can easily be converted into a sublattice model using the following relations:

$$G^{\phi}_{\text{Me:H}} = G^{\phi}_{\text{H}}; \qquad {}^{v}L^{\phi}_{\text{Me:H,Va}} = {}^{v}L^{\phi}_{\text{Me,H}} \tag{4}$$

Hydrogen occupies both the tetrahedral and octahedral interstitial sites in b.c.c.- and octahedral sites in f.c.c. iron [71]. The crystallographic sites occupied by H in the allotropic forms of Sm: (α-Sm), (β-Sm), and (γ-Sm) were not experimentally identified. The b.c.c., f.c.c., h.c.p., and rhombohedral (α-Sm) phases were described by a two-sublattice model, with one sublattice fully occupied by Fe or Sm and the other one for the interstitial species (H or vacancy). For one mole of formula unit $(\text{Fe,Sm})_1(\text{H,Va})_1$ the expression for the Gibbs energy is given by Eq. (3), where ϕ stands for b.c.c., f.c.c., h.c.p., or rhombohedral (α-Sm) phase. $G^{\phi}_{\text{Me:Va}}$ represents the Gibbs energy of the corresponding polymorphic modification of pure Fe or Sm and $G^{\phi}_{\text{Me:H}}$ is the Gibbs energy of hypothetical compounds MeH with the b.c.c., f.c.c., h.c.p., or rhombohedral structure of α-Sm, i.e. when all the interstitial sites in the lattice are filled with hydrogen. To avoid the possible interference with $\text{SmH}_{2\pm x}$, a limit of 1.0 was set for the number of

interstitial sites in the thermodynamic modeling of interstitial solid solution phases, which allows the experimental homogeneity range being reproduced by calculations. It would be also possible to use models in which hydrogen occupies all octahedral and/or tetrahedral voids in the given type of structure. For example, the limiting composition for b.c.c. phase would be FeH_9! However, in such a case, extremely large excess Gibbs energies would be required to fit the experimental data. Furthermore, additional complication arises due to the site blocking at high H-concentrations (i.e. when the occupation of an interstitial site is prevented by the prior occupation of a neighboring interstitial site, since the effective repulsion exists between near-neighbor H-atoms). Site blocking gives rise to the non-ideal entropy of mixing on the interstitial sublattice that is not compatible with the models commonly used.

In the rhombohedral structure of $Fe_{17}Sm_2$, there are two different interstitial sites, which are likely to accommodate interstitial atoms: a large octahedral site 9*e* and a smaller tetrahedral site 18*g* [2]. Hydrogen occupies both 9*e* and 18*g* sites. It has been found that 9*e* site is completely occupied by three H, but the 18*g* occupancy is restricted to two hydrogen atoms per formula unit $Fe_{17}Sm_2$ [2]. This behaviour reflects a repulsive short-range interaction between different hydrogen atoms. To preserve the ideal entropy of mixing (see above) two interstitial sublattices were combined, fixing the maximum possible occupation of interstitials to five hydrogen atoms per formula unit. The thermodynamic model for $Fe_{17}Sm_2H_\delta$ includes three sublattices with the first and the second sublattices fully occupied by samarium and iron atoms, respectively, and the third sublattice containing hydrogens and vacancies, i.e. $(Sm)_2(Fe)_{17}(H,Va)_5$. With only vacancies in the third sublattice this model reduces to the binary stoichiometric compound $Fe_{17}Sm_2$. The Gibbs energy of $Fe_{17}Sm_2H_\delta$ referred to 1 mol of metal atoms[1] is then expressed as:

$$G^{2:17:5}_{Sm_{2/19}:Fe_{17/19}:(H,Va)_{5/19}} = y_{Va}G^{2:17:5}_{Sm:Fe:Va} + y_H G^{2:17:5}_{Sm:Fe:H} + \frac{5}{19}RT(y_H \ln y_H + y_{Va} \ln y_{Va}) + y_H y_{Va} L^{2:17:5}_{Sm:Fe:H,Va} \quad (5)$$

where $G^{2:17:5}_{Sm:Fe:Va}$ is the molar Gibbs energy of $Fe_{17}Sm_2$, $G^{2:17:5}_{Sm:Fe:H}$ is the Gibbs energy of the compound $Fe_{17}Sm_2H_5$ and $L^{2:17:5}_{Sm:Fe:H,Va}$ is the interaction parameter of hydrogen atoms and vacancies in the interstitial sublattice, all referred to 1 mol of metal atoms.

The samarium dihydride phase exists with both deficiency and excess of hydrogen with respect to the ideal stoichiometric composition SmH_2 [46]. The thermodynamic model for $SmH_{2\pm x}$ phase was selected on the basis of crystallographic information (see above) and includes three sublattices, with one sublattice fully occupied by Sm atoms and two interstitial sublattices representing tetrahedral and octahedral sites, respectively, where hydrogen atoms and vacancies mix randomly, i.e. $(Sm)_1(H,Va)_2(H,Va)_1$. Using the notations y'_i and y''_i for the site fraction of hydrogen or vacancies in the first and second interstitial sublattices, respectively, this model yields the following expression for the Gibbs energy of the non-stoichiometric samarium dihydride (referred to 1 mol of formula unit):

$$\begin{aligned} G^{SmH_2}_{Sm:(H,Va)_2:(H,Va)} &= y'_{Va}y''_{Va}G^{SmH_2}_{Sm:Va:Va} + y'_H y''_{Va}G^{SmH_2}_{Sm:H:Va} \\ &\quad + y'_{Va}y''_H G^{SmH_2}_{Sm:Va:H} + y'_H y''_H G^{SmH_2}_{Sm:H:H} \\ &\quad + G^{SmH_2}_{id} + G^{SmH_2}_{ex} \\ G^{SmH_2}_{id} &= RT(2y'_H \ln y'_H + 2y'_{Va} \ln y'_{Va} + y''_H \ln y''_H \\ &\quad + y''_{Va} \ln y''_{Va}) \\ G^{SmH_2}_{ex} &= y'_{Va}y''_H y''_{Va} L^{SmH_2}_{Sm:Va:H,Va} + y'_H y''_H y''_{Va} L^{SmH_2}_{Sm:H:H,Va} \\ &\quad + y'_H y'_{Va} y''_{Va} L^{SmH_2}_{Sm:H,Va:Va} + y'_H y'_{Va} y''_H L^{SmH_2}_{Sm:H,Va:H} \end{aligned} \quad (6)$$

where $G^{SmH_2}_{Sm:H:Va}$ represents the Gibbs energy of the stoichiometric SmH_2 phase. $G^{SmH_2}_{Sm:Va:Va}$ and $G^{SmH_2}_{Sm:H:H}$ are the Gibbs energies of the hypothetical pure Sm and SmH_3, respectively, with the f.c.c. crystal structure, while $G^{SmH_2}_{Sm:Va:H}$ stands for the Gibbs energy of the isostructural fictitious compound SmH, in which all the octahedral sites are fully occupied by hydrogen atoms and all the tetrahedral sites are vacant. The parameters $L^{SmH_2}_{Sm:H,Va:H}$, $L^{SmH_2}_{Sm:H,Va:Va}$, $L^{SmH_2}_{Sm:H:H,Va}$, and $L^{SmH_2}_{Sm:Va:H,Va}$ represent the interaction energy between hydrogen and vacancies in the second and third sublattices, respectively.

The samarium trihydride can be obtained either stoichiometrically or with hydrogen deficiency. Although two types of hexagonal structure were reported for this phase, little is known about this polymorphism and no distinct phase boundary has been determined. Therefore, in the thermodynamic modeling only one trihydride phase SmH_{3-y} was taken into account. In the structure of HoH_3-type [73], hydrogens are located in three different crystallographic sites but nothing is known about, where the vacancies can be formed. To simplify the thermodynamic model, a three-sublattice model was employed with the first and the second sublattices fully occupied by Sm and hydrogen atoms, respectively, and the third sublattice containing hydrogens and vacancies, i.e. $(Sm)_1(H)_2(H,Va)_1$. The Gibbs energy of SmH_{3-y} referred to 1 mol of formula unit is then expressed as:

$$G^{SmH_3}_{Sm:H_2:(H,Va)} = y_{Va}G^{SmH_3}_{Sm:H:Va} + y_H G^{SmH_3}_{Sm:H:H} + RT(y_H \ln y_H + y_{Va} \ln y_{Va}) + y_H y_{Va} L^{SmH_3}_{Sm:H:H,Va} \quad (7)$$

where $G^{SmH_3}_{Sm:H:H}$ and $G^{SmH_3}_{Sm:H:Va}$ are the Gibbs energies of the

[1]The number of sites must be unity to calculate the magnetic contribution to the Gibbs energy correctly [72] (see also below).

stoichiometric SmH_3 phase and of the hypothetical hexagonal samarium dihydride, respectively. $L^{SmH_3}_{Sm:H:H,Va}$ represents the interaction energy between hydrogen and vacancies in the third sublattice.

The compounds $Fe_{17}Sm_2$, Fe_3Sm, and Fe_2Sm were assumed to be stoichiometric. Their Gibbs energy is expressed by Eq. (1). The description of $Fe_{17}Sm_2$ compound is a part of the sublattice model for the interstitial solid solution phase $Fe_{17}Sm_2H_\delta$. Although a homogeneity range for $Fe_{17}Sm_2$, Fe_3Sm, and Fe_2Sm was reported [45], there is insufficient crystallographic data to use the sublattice model here.

Magnetic ordering in b.c.c. Fe, f.c.c. Fe, $Fe_{17}Sm_2H_\delta$, Fe_3Sm, and Fe_2Sm results in an additional contribution to the Gibbs energy. For a phase with the magnetic moment β (Bohr's magnetons per mol of atoms) and the magnetic ordering (Curie or Neel) temperature T_C (K) the magnetic contribution to its Gibbs energy is expressed as [74,75]:

$$G^{mag} = RT \cdot \ln(\beta + 1) \cdot f(T/T_C) \tag{8}$$

where $f(T/T_C)$ represents the polynomial function [75]. For the interstitial solid solution phases T_C and β are composition-dependent:

$$(T_C)^{\phi}_{Me:(H,Va)} = y_{Va}(T_C)^{\phi}_{Me:Va} + y_H(T_C)^{\phi}_{Me:H} + y_H y_{Va}(T_C)^{\phi}_{Me:H,Va} \tag{9}$$

$$\beta^{\phi}_{Me:(H,Va)} = y_{Va}\beta^{\phi}_{Me:Va} + y_H\beta^{\phi}_{Me:H} + y_H y_{Va}\beta^{\phi}_{Me:H,Va} \tag{10}$$

where the quantities $(T_C)^{\phi}_{Me:Va}$ and $\beta^{\phi}_{Me:Va}$ correspond to pure Fe or $Fe_{17}Sm_2$, while $(T_C)^{\phi}_{Me:H}$ and $\beta^{\phi}_{Me:H}$ refer to the fully hydrogenated compounds FeH or $Fe_{17}Sm_2H_5$. $(T_C)^{\phi}_{Me:H,Va}$ and $\beta^{\phi}_{Me:H,Va}$ account for the deviation in the magnetic properties of a solution from the linear interpolation between end-members.

5. Optimization procedure

5.1. General remarks

The thermodynamic model parameters were evaluated on the basis of the available experimental information using the optimization module PARROT of the multicomponent thermodynamic software TC4A (Thermo-Calc for Academic) [76,77]. A general rule for selection of the adjustable parameters is that only those coefficients that are determined by the experimental values should be adjusted [78]. The thermodynamic description of the vapor phase contains no adjustable parameters. Unlike alloys, the metal–hydrogen systems are characterized by relatively large negative entropy for solution of H in a metal and for the formation of hydrides [79]. The latter includes as a major part the entropy difference of about 130 $J\,mol^{-1}\,K^{-1}$ due to the dissociation of the H_2 molecule and incorporation of two hydrogen atoms into the crystal lattice [80]. The entropy for solution of H in a molten metal is also negative, but its absolute value is smaller than that for solid phases because only short-range order exists in the liquid state. Thus, the Gibbs energy terms of any condensed phase containing hydrogen (Eqs. (3), (5)–(7)) without the reference part[2] should be represented by Eq. (1) with at least two non-zero parameters a and $b>0$, which stand for ΔH and $-\Delta S$ for hydrogen solution or hydride formation.

Individual data points of both the phase diagram type and the thermodynamic type (e.g. equilibrium hydrogen pressure) were used together with their associated experimental errors and stored in a file for the optimization procedure. But, in fact, the experimental data from different sources are often not consistent and even cannot be reconciled with each other in some cases. If contradictions between different experimental data sets became apparent, their individual reliability was checked and a certain weight was assigned to each piece of information. The thermodynamic parameters resulted from the least-squares optimization are given in Table 1.

5.2. The Fe–Sm system

First of all, the magnetic parameters for $Fe_{17}Sm_2$, Fe_3Sm, and Fe_2Sm were assessed (Table 2). With such selection of data, the standard deviation for the assessed values of the Curie temperature and magnetic moment does not exceed ± 10 K and ± 0.04 μ_B, respectively.

The thermodynamic description of the Fe–Sm system was then evaluated using the phase diagram data [12], the heat capacity and entropy of $Fe_{17}Sm_2$ measured in the present work, and the temperatures of the invariant recombination reaction in the system $Fe_{17}Sm_2$–hydrogen gas. The thermodynamic parameters given in Ref. [10] were used as start values in the optimization. For the liquid phase, two interaction parameters could be adjusted to fit the measured temperatures of the liquidus and those of the invariant reactions. The lattice stability of the hypothetical f.c.c. samarium and of iron with the rhombohedral structure of α-Sm was approximated as $+5$ $kJ\,mol^{-1}$. To account for the extremely low terminal solubilities, the interaction parameters for b.c.c., f.c.c., h.c.p., and rhombohedral α-Sm phase were fixed to $+1$ $MJ\,mol^{-1}$. For the intermetallic compounds Fe_3Sm and Fe_2Sm, no information exists other than they are stable and take part in two peritectic and one eutectic reaction. Therefore, their Gibbs energy of formation was represented using temperature-independent enthalpy and entropy terms. In the case of $Fe_{17}Sm_2$ more experimental information is available (see above), so that the thermodynamic description of this phase contains coefficients a to f in Eq. (1).

[2]The reference part stands for the Gibbs energy of the mechanical mixture of pure elements in a corresponding proportion (see also Table 1).

Table 1
Summary of the thermodynamic parameters describing the Fe–Sm, Fe–H, H–Sm, and $Fe_{17}Sm_2$–H_2 systems referred to stable element reference H^{SER} (T=298.15 K, P=101 325 Pa)

Parameter	Eq.	T_{max}	Ref.
${}^{\circ}G^{gas}_{H_2} = -9522.9741 + 78.5273879T - 31.35707T \ln T$	(2)	1000	[67]
$+ 0.0027589925T^2 - 7.46390667E{-}7T^3 + 56582.3T^{-1}$;			
$+ 180.108664 - 15.6128256T - 17.84857T \ln T$		2100	
$- 0.00584168T^2 + 3.14618667E{-}7T^3 - 1280036T^{-1}$;			
$- 18840.1663 + 92.3120255T - 32.05082T \ln T$		6000	
$- 0.0010728235T^2 + 1.14281783E{-}8T^3 + 3561002.5T^{-1}$;			
${}^{\circ}G^{gas}_{H} = +211801.621 + 24.4989821T - 20.78611T \ln T$;	(2)	6000	[67]
${}^{\circ}G^{gas}_{Fe} = +405563.032 + 35.536443T - 32.8261T \ln T$	(2)	900	[67]
$+ 0.00908265T^2 - 1.34845667E{-}6T^3 + 108791.4T^{-1}$;			
$+ 414134.539 - 53.8401478T - 19.84276T \ln T$		2400	
$+ 6.959445E{-}5T^2 - 1.30682983E{-}7T^3 - 976411.5T^{-1}$;			
$+ 410389.799 - 49.3269727T - 20.12513T \ln T$		5500	
$- 5.66549E{-}4T^2 - 5.290265E{-}8T^3 + 887592T^{-1}$;			
$+ 521855.538 - 375.147472T + 18.70844T \ln T$		10 000	
$- 0.00634452T^2 + 1.038655E{-}7T^3 - 55487750T^{-1}$;			
${}^{\circ}G^{gas}_{Sm} = +197389.067 + 1.48784782T - 27.25423T \ln T$	(2)	1100	[67]
$- 0.004541758T^2 + 9.66538167E{-}7T^3 - 40312.025T^{-1}$;			
$+ 169303.384 + 252.985232T - 62.81893T \ln T$		2800	
$+ 0.01571714T^2 - 1.22936017E{-}6T^3 + 4157452T^{-1}$;			
$+ 670735.118 - 1795.71893T + 193.8858T \ln T$		4400	
$- 0.042934865T^2 + 1.295334E{-}6T^3 - 1.80147E8T^{-1}$;			
$- 295842.195 + 803.309978T - 113.2675T \ln T$		5900	
$+ 2.614958E{-}4T^2 + 1.61802217E{-}7T^3 + 3.8803995E8T^{-1}$;			
$- 658799.787 + 1797.39061T - 230.6297T \ln T$		9000	
$+ 0.016324745T^2 - 2.371995E{-}7T^3 + 5.82118E8T^{-1}$;			
$+ 315946.24 + 222.042246T - 56.247T \ln T$		10 000	
$+ 0.0024534215T^2 - 3.03084333E{-}8T^3 - 4.404249E8T^{-1}$;			
Liquid $(Fe,Sm)_1(H,Va)_1$			
$G^{liq.}_{Fe:Va} = +12040.17 - 6.55843T - 3.67516E{-}21T^7 + G^{b.c.c.}_{Fe:Va}$;	(3)	1811	[64]
$- 10838.83 + 291.302T - 46T \ln T$;		6000	
$G^{liq.}_{Fe:H} = +28172.583 + 40.0041662T + 0.5{}^{\circ}G^{gas}_{H_2} + G^{liq.}_{Fe:Va}$;	(3)	6000	This work
$G^{liq.}_{Sm:Va} = +3468.783 + 20.117456T - 11.6968284T \ln T$	(3)	1190	[64]
$- 0.032418177T^2 + 4.544272E{-}6T^3 + 23528T^{-1}$;			
$- 11728.229 + 273.487076T - 50.208T \ln T$;		1345	
$+ 11327.85 - 8.707299T + G^{\alpha\text{-}Sm}_{Sm:Va}$;		2100	
$G^{liq.}_{Sm:H} = -18701.1193 + 12T + 0.5{}^{\circ}G^{gas}_{H_2} + G^{liq.}_{Sm:Va}$;	(3)	2100	This work
$L^{liq.}_{Fe,Sm:Va} = L^{liq.}_{Fe,Sm:H} = +72.5493721 - 0.876430441T$;	(3)	6000	This work
$L^{liq.}_{Fe,Sm:Va} = L^{liq.}_{Fe,Sm:H} = -17962.8286 + 21.1635471T$;	(3)	6000	This work
$L^{liq.}_{Sm:H,Va} = -811.476845$;	(3)	2100	This work
α-Sm $(Fe,Sm)_1(H,Va)_1$			
$G^{\alpha\text{-}Sm}_{Sm:Va} = -3872.013 - 32.10748T - 1.6485T \ln T - 0.050254T^2$	(3)	700	[64]
$+ 1.010345E{-}5T^3 - 82168T^{-1}$;			
$- 50078.215 + 627.869894T - 102.665T \ln T$		1190	
$+ 0.0474522T^2 - 7.538383E{-}6T^3 + 3861770T^{-1}$;			
$+ 289719.819 - 2744.50976T + 381.41982T \ln T$		1345	
$- 0.2549863 38T^2 + 2.7512152E{-}5T^3 - 40102102T^{-1}$;			
$- 23056.079 + 282.194375T - 50.208T \ln T$;		2100	
$G^{\alpha\text{-}Sm}_{Sm:H} = -78732.8080 + 53.5159713T + G^{\alpha\text{-}Sm}_{Sm:Va} + 0.5{}^{\circ}G^{gas}_{H_2}$;	(3)	2100	This work
$G^{\alpha\text{-}Sm}_{Fe:Va} = +5000 + G^{b.c.c.}_{Fe:Va}$;	(3)	6000	This work
$G^{\alpha\text{-}Sm}_{Fe:H} = +39000 + 42.7T + G^{b.c.c.}_{Fe:Va} + 0.5{}^{\circ}G^{gas}_{H_2}$;	(3)	6000	This work
$L^{(\alpha\text{-}Sm)}_{Fe,Sm:Va} = L^{(\alpha\text{-}Sm)}_{Fe,Sm:H} = +1.0E6$;	(3)	6000	This work
$L^{(\alpha\text{-}Sm)}_{Sm:H,Va} = L^{(\alpha\text{-}Sm)}_{Fe:H,Va} = +3457.14915$;	(3)	2100	This work
h.c.p. $(Fe,Sm)_1(H,Va)_1$			
$G^{h.c.p.}_{Fe:Va} = -3705.78 + 12.591T - 1.15T \ln T + 6.4E{-}4T^2 + G^{b.c.c.}_{Fe:Va}$;	(3)	1811	[64]
$- 3957.199 + 5.24951T + 4.9251E30T^{-9} + G^{b.c.c.}_{Fe:Va}$;	(3)	6000	
$G^{h.c.p.}_{Fe:H} = +34000 + 42.7T + G^{h.c.p.}_{Fe:Va} + 0.5{}^{\circ}G^{gas}_{H_2}$;	(3)	6000	This work
$G^{h.c.p.}_{Sm:Va} = +69.9774 - 0.069491T + G^{\alpha\text{-}Sm}_{Sm:Va}$;	(3)	2100	[66]
$G^{h.c.p.}_{Sm:H} = -104354.3426 + 71.7812162T + G^{\alpha\text{-}Sm}_{Sm:Va} + 0.5{}^{\circ}G^{gas}_{H_2}$;	(3)	2100	This work
$L^{h.c.p.}_{Fe,Sm:Va} = L^{h.c.p.}_{Fe,Sm:H} = +1.0E6$;	(3)	6000	This work

Table 1. Continued

Parameter	Eq.	T_{max}	Ref.
$L^{h.c.p.}_{Sm:H,Va} = L^{h.c.p.}_{Fe:H,Va} = +32203.1096 - 21.585983T$;	(3)	2100	This work
b.c.c. $(Fe,Sm)_1(H,Va)_1$			
$G^{b.c.c.}_{Fe:Va} = +1225.7 + 124.134T - 23.5143T \ln T - 0.00439752T^2$	(3)	1811	[64]
$- 5.8927E{-}8T^3 + 77359T^{-1}$;			
$- 25383.581 + 299.31255T - 46T \ln T + 2.29603E31T^{-9}$;		6000	
$G^{b.c.c.}_{Fe:H} = +71098.3995 - 986.034062T + 155.942278T \ln T$	(3)	1811	This work
$- 0.0894166963T^2 + G^{b.c.c.}_{Fe:Va} + 0.5°G^{gas}_{H_2}$;			
$+ 80393.95 + 16.72144T$;	(3)	6000	
$G^{b.c.c.}_{Sm:Va} = -4368.72 + 55.972523T - 16.9298494T \ln T$	(3)	1190	[64]
$- 0.025446016T^2 + 3.579527E{-}6T^3 + 94209T^{-1}$;			
$- 15957.862 + 253.121044T - 46.9445T \ln T$;	(3)	1345	
$+ 111191.653 - 624.680805T + 71.6856914T \ln T$	(3)	2100	
$- 0.047314968T^2 + 3.329865E{-}6T^3 - 24870276T^{-1}$;			
$G^{b.c.c.}_{Sm:H} = -87329.5649 + 57.9009778T + G^{b.c.c.}_{Sm:Va} + 0.5°G^{gas}_{H_2}$;	(3)	2100	This work
$L^{b.c.c.}_{Fe,Sm:Va} = L^{b.c.c.}_{Fe,Sm:H} = +1.0E6$;	(3)	6000	This work
$L^{b.c.c.}_{Sm:H,Va} = +6061.9523$;	(3)	2100	This work
$(T_C)^{b.c.c.}_{Fe:Va} = 1043$; $\beta^{b.c.c.}_{Fe:Va} = 2.22$;	(8)–(10)	6000	[64]
$(T_C)^{b.c.c.}_{Fe:H} = 0$; $\beta^{b.c.c.}_{Fe:H} = 0$;	(8)–(10)	6000	This work
f.c.c. $(Fe,Sm)_1(H,Va)_1$			
$G^{f.c.c.}_{Fe:Va} = -1462.4 + 8.282T - 1.15T \ln T + 6.4E{-}4T^2 + G^{b.c.c.}_{Fe:Va}$;	(3)	1811	[64]
$- 1713.815 + 0.94001T + 4.9251E30T^{-9} + G^{b.c.c.}_{Fe:Va}$;		6000	
$G^{f.c.c.}_{Fe:H} = +25746.7209 + 48.1250165T + G^{f.c.c.}_{Fe:Va} + 0.5°G^{gas}_{H_2}$;	(3)	6000	This work
$G^{f.c.c.}_{Sm:Va} = +5000 + G^{\alpha\text{-Sm}}_{Sm:Va}$;	(3)	2100	This work
$G^{f.c.c.}_{Sm:H} = +39000 + 42.7T + G^{\alpha\text{-Sm}}_{Sm:Va} + 0.5°G^{gas}_{H_2}$;	(3)	2100	This work
$L^{f.c.c.}_{Fe,Sm:Va} = L^{f.c.c.}_{Fe,Sm:H} = +1.0E6$;	(3)	6000	This work
$(T_C)^{f.c.c.}_{Fe:Va} = 67$; $\beta^{f.c.c.}_{Fe:Va} = 0.7$;	(8)–(10)	6000	[64]
$(T_C)^{f.c.c.}_{Fe:H} = 0$; $\beta^{f.c.c.}_{Fe:H} = 0$;	(8)–(10)	6000	This work
SmH_2 $(Sm)_1(H,Va)_2(H,Va)_1$			
$G^{SmH_2}_{Sm:H:Va} = -194994.854 + 131.551681T + G^{\alpha\text{-Sm}}_{Sm:Va} + °G^{gas}_{H_2}$;	(6)	2100	This work
$G^{SmH_2}_{Sm:H:H} = -228089.27 + 187.223575T + G^{\alpha\text{-Sm}}_{Sm:Va} + 1.5°G^{gas}_{H_2}$;	(6)	2100	This work
$G^{SmH_2}_{Sm:Va:Va} = +141138.908 + G^{\alpha\text{-Sm}}_{Sm:Va}$;	(6)	2100	This work
$G^{SmH_2}_{Sm:Va:H} = +141138.908 + G^{\alpha\text{-Sm}}_{Sm:Va} + 0.5°G^{gas}_{H_2}$;	(6)	2100	This work
$L^{SmH_2}_{Sm:Va:H,Va} = L^{SmH_2}_{Sm:H:H,Va} = -2671.83177 - 5.60912667T$;	(6)	2100	This work
$L^{SmH_2}_{Sm:H,Va:Va} = L^{SmH_2}_{Sm:H,Va:H} = -167331.063$;	(6)	2100	This work
SmH_3 $(Sm)_1(H)_2(H,Va)_1$			
$G^{SmH_3}_{Sm:H:H} = -229287.977 + 185.830052T + G^{\alpha\text{-Sm}}_{Sm:Va} + 1.5°G^{gas}_{H_2}$;	(7)	2100	This work
$G^{SmH_3}_{Sm:H:Va} = -186090.838 + 126.795384T + G^{\alpha\text{-Sm}}_{Sm:Va} + °G^{gas}_{H_2}$;	(7)	2100	This work
$L^{SmH_3}_{Sm:H:H,Va} = -10297.5795$;	(7)	2100	This work
2:17:5 $(Sm)_{2/19}(Fe)_{17/19}(H,Va)_{5/19}$			
$G^{2:17:5}_{Sm:Fe:Va} = -3287.51224 - 56.6232245T + 8.40715399T \ln T$	(5)	6000	This work
$- 2.59607031E-3T^2 - 2.08252746E{-}7T^3 - 94492.7287T^{-1}$			
$+ 0.89473684G^{b.c.c.}_{Fe:Va} + 0.10526316G^{\alpha\text{-Sm}}_{Sm:Va}$;			
$G^{2:17:5}_{Sm:Fe:H} = -14175.2698 + 14.7387267T + 0.89473684G^{b.c.c.}_{Fe:Va}$	(5)	6000	This work
$+ 0.10526316G^{\alpha\text{-Sm}}_{Sm:Va} + 0.13157895°G^{gas}_{H_2}$;			
$L^{2:17:5}_{Sm:Fe:H,Va} = -6360.88134 + 9.20283816T$;	(5)	6000	This work
$(T_C)^{2:17:5}_{Me:Va} = 394$; $\beta^{2:17:5}_{Me:Va} = 1.83$;	(8)–(10)	6000	This work
$(T_C)^{2:17:5}_{Me:H} = 553$; $\beta^{2:17:5}_{Me:H} = 2.09$;	(8)–(10)	6000	[19]
Fe_3Sm $(Fe)_{0.75}(Sm)_{0.25}$			
$G^{Fe_3Sm} = -6600.8005 + 1.73807443T + 0.75G^{b.c.c.}_{Fe:Va} + 0.25G^{\alpha\text{-Sm}}_{Sm:Va}$;	(1)	6000	This work
$(T_C)^{Fe_3Sm} = 653$; $\beta^{Fe_3Sm} = 1.31$;	(8)	6000	This work
Fe_2Sm $(Fe)_{2/3}(Sm)_{1/3}$			
$G^{Fe_2Sm} = -5903.29027 + 1.11094574T + 2/3G^{b.c.c.}_{Fe:Va} + 1/3G^{\alpha\text{-Sm}}_{Sm:Va}$;	(1)	6000	This work
$(T_C)^{Fe_2Sm} = 682$; $\beta^{Fe_2Sm} = 0.97$;	(8)	6000	This work

Values are given in SI units (Joule, mol, and Kelvin), except β (μ_B per mole of atoms). The interaction parameters, which are not specified in the table equal zero by default. 'Eq.' indicates the equation number in the text.

Table 2
Magnetic properties of the intermetallic compounds in the Fe–Sm system

$Fe_{17}Sm_2$				Fe_3Sm				Fe_2Sm			
T_C	β	Ref.	Use[a]	T_C	β	Ref.	Use[a]	T_C	β	Ref.	Use[a]
385	–	[12]	+	650	–	[12]	+	675	–	[12]	+
363	1.868[b]	[13]	+	655	1.330[c]	[35]	+	683	0.801[d]	[39]	−
395	–	[14]	+	663	1.147[d]	[36]	−	698	0.939[d]	[37]	−
412	–	[15]	+	683	1.139[d]	[37]	−	692	0.967[b]	[40]	+
389	–	[16]	+	644	1.280[b]	[38]	+	677	0.933[c]	[41]	+
390	–	[17]	+					669	1.008[b]	[42]	+
410	–	[18]	+								
408	1.367[d]	[19]	−								
391	–	[20]	+								
387	–	[21]	+								
412	1.666[b]	[22]	−								
412	1.829[b]	[23]	+								
385	–	[24]	+								
387	–	[25]	+								
391	–	[26]	+								
389	1.795[b]	[27]	+								
389	–	[28]	+								
400	–	[29]	+								
398	–	[30]	+								
390	–	[31]	+								
390	–	[32]	+								
387	1.826[b]	[33]	+								
398	–	[34]	+								
394± 10	1.83± 0.03	This work, assessed		653± 8	1.31± 0.04	This work, assessed		682± 10	0.97± 0.04	This work, assessed	

T_C and β stand for Curie temperature (K) and a mean magnetic moment (μ_B per mol of atoms), respectively.
[a] This column indicates whether the values were used (+) or not used (−) in the optimisation.
[b] Measured at 4.2 K.
[c] Extrapolated to $T=0$ K.
[d] Measured at room temperature.

The experimental information for Fe–Sm system is summarized in Table 3. The measured quantity given after data description in Table 3 indicates what had to be reproduced with the best fit of the thermodynamic parameters. Other values reported for a given type of experiment were used as experimental conditions. The criterion to distinguish between the measured values and the conditions was the description of experiment provided in the original publication.

The temperature of the invariant recombination reaction in $Fe_{17}Sm_2$–H_2 system at 60 kPa hydrogen pressure was not used, because it was found to be incompatible with the temperature of the peritectic reaction: Liquid + f.c.c. Fe ⇒ $Fe_{17}Sm_2$. The optimization was successful using the value for $pH_2=101\,325$ Pa instead [61]. Furthermore, the heat capacity data for $Fe_{17}Sm_2$ could be used in a limited range of temperature only ($T>270$ K), since the reliable data for the Gibbs energy of pure α-Sm below room temperature are still lacking.

5.3. The H–Sm system

Table 4, which is constructed like Table 3 gives a summary of the experimental data for the H–Sm system. Those, which show disagreement or inconsistency with the most data sets have not been included in the optimization. The liquid, (γ-Sm), and (α-Sm) phases were modeled with the temperature-independent interaction parameters $L^{\text{liq.}}_{\text{Sm:H,Va}}$, $L^{\text{b.c.c.}}_{\text{Sm:H,Va}}$, and $L^{(\alpha\text{-Sm})}_{\text{Sm:H,Va}}$, respectively. At the same time, to take into account an extensive amount of the experimental data for (β-Sm) phase, the interaction parameter $L^{\text{h.c.p.}}_{\text{Sm:H,Va}}$ was optimized as a linear function of temperature.

The thermodynamic description of the $SmH_{2\pm x}$ phase contains four Gibbs energy terms and four interaction parameters (Eq. (6)). However, some of them refer to the fictive compounds, so that constraints should have been applied to obtain the converging solution in the optimization. Only two Gibbs energy terms $G^{\text{SmH}_2}_{\text{Sm:H:Va}}$ and $G^{\text{SmH}_2}_{\text{Sm:H:H}}$ were described using two non-zero coefficients a and b in Eq. (1), which could be adjusted. The Gibbs energy terms for the fictive phases, i.e. $G^{\text{SmH}_2}_{\text{Sm:Va:Va}}$ and $G^{\text{SmH}_2}_{\text{Sm:Va:H}}$ were represented by single coefficient a plus the corresponding reference part (see Table 1). The interaction parameters for H and Va in the specific sublattice were set equal irrespective of the constituent in the other one (i.e. $L^{\text{SmH}_2}_{\text{Sm:H,Va:Va}}=L^{\text{SmH}_2}_{\text{Sm:H,Va:H}}$ and $L^{\text{SmH}_2}_{\text{Sm:Va:H,Va}}=L^{\text{SmH}_2}_{\text{Sm:H:H,Va}}$). Following the idea employed in the expression of the Gibbs

Table 3
Selection of experimental information for the Fe–Sm system

Data description	Experimental method	Measured quantity	Use[a]	Ref.
Peritectic reaction: Liquid + f.c.c. Fe ⇒ $Fe_{17}Sm_2$	DTA	T	+	[12]
Peritectic reaction: Liquid + $Fe_{17}Sm_2$ ⇒ Fe_3Sm	DTA	T	+	[12]
Peritectic reaction: Liquid + Fe_3Sm ⇒ Fe_2Sm	DTA	T	+	[12]
Eutectic reaction: Liquid ⇒ Fe_2Sm + α-Sm	DTA	T	+	[12]
Liquidus	DTA	T at given x_{Sm}	+	[12]
Entropy of $Fe_{17}Sm_2$	Calorimetry	°S at $T = 298.15$ K	+	This work
Heat capacity of $Fe_{17}Sm_2$	Calorimetry	C_p at given T ($T > 270$ K)	+	This work
Invariant reaction: Fe + $SmH_{2\pm x}$ ⇒ $Fe_{17}Sm_2 + H_2$	Monitoring gas flow rate	T at given P	+	[61]
	TPA[b]	($P = 2.5$–12 kPa)	+	[63]
		($P = 60$ kPa)	−	[63]

[a] This column indicates whether the values were used (+) or not used (−) in the optimization.
[b] Thermopiezic analysis.

energy terms for $SmH_{2\pm x}$ the linear temperature dependence was introduced only for the interactions in the third sublattice. The interactions in the second sublattice, which is almost always filled with hydrogens were assumed temperature-independent.

In the case of the SmH_{3-y} phase, the linear dependence on the temperature was set for both Gibbs energy terms $G^{SmH_3}_{Sm:H:H}$ and $G^{SmH_3}_{Sm:H:Va}$. The homogeneity range of SmH_{3-y} is narrow and does not change significantly with the temperature [46], so that no temperature dependence was assumed for the interaction parameter $L^{SmH_3}_{Sm:H:H,Va}$.

The optimization was performed stepwise using, respectively, zero and 130 J/(K^{-1} mol^{-1} H_2) as start values for the enthalpy and entropy of formation of the hypothetical compounds SmH with the rhombohedral (α-Sm), h.c.p., and b.c.c. structure. Those for samarium dihydride phase were the heat and entropy of formation of the stoichiometric compound SmH_2 calculated from the equilibrium hydrogen pressures [83]. First, the parameters for (β-Sm) and $SmH_{2\pm x}$ phases were determined using corresponding equilibrium hydrogen pressures and phase diagram data. Then, the optimization has been continued by successively taking into account (α-Sm), (γ-Sm), the liquid phase, and SmH_{3-y}. For each step, the parameters obtained previously were changed a little to fit the new experimental information. Finally, the whole system was optimized using all the selected experimental data simultaneously.

5.4. The Fe–H system

Since the hydrogen solubility in solid iron is very small within the pressure range of interest ($P < 10$ MPa) the interaction between hydrogen atoms and vacancies in the second sublattice of the b.c.c. and f.c.c. phases was neglected. The interaction parameters $L^{b.c.c.}_{Fe:H,Va}$ and $L^{f.c.c.}_{Fe:H,Va}$ in Eq. (3) were therefore fixed to zero. Also the magnetic parameters for b.c.c. and f.c.c. phase were taken to be the same as for pure iron, assuming that no magnetic ordering exists in the hypothetical compounds FeH. The Gibbs energy terms were described using the coefficients a to d ($G^{b.c.c.}_{Fe:H}$) and a to b ($G^{f.c.c.}_{Fe:H}$) in Eq. (1) to fit the solubility data, which are available over the extended temperature region. For the liquid phase, the experimental information is limited to a few points on the liquid/gas phase boundary that made possible the adjustment of $G^{liq.}_{Fe:H}$ as linear function of temperature with zero interaction parameter $L^{liq.}_{Fe:H,Va}$.

Table 5, which is again constructed similarly to Table 3 gives a summary of the experimental data for the Fe–H system. The data for hydrogen solubility in b.c.c. Fe at 101 325 Pa pressure used in the present optimization were selected on the basis of the latest compilation of Kiuchi and McLellan [50], because it had excluded all the data that might be influenced by surface-related effects. In the f.c.c. region, there are several measurements [53,92–95], which fall into the same range: they were used as a reference in the selection of data to be used in the optimization. From the solubility data at higher pressure than the atmospheric one, those were selected which conform to Sieverts law, since it is implied in the model formulations. The optimization was performed stepwise using, respectively, zero and 130 J/(K^{-1} mol^{-1} H_2) as start values for the enthalpy and entropy of formation of the hypothetical compounds FeH with the b.c.c. and f.c.c.

Table 4
Selection of experimental information for the H–Sm system

Data description	Exp. method	Measured quantity	Use[a]	Ref.
Eutectoid reaction: (β-Sm)⇒(α-Sm)+SmH_{2-x}	DTA	T at given P	+	[81]
Gas–peritectoid reaction: $SmH_{2\pm x}+H_2 \Rightarrow SmH_{3-y}$	DTA	T at given P	+	[47]
Enthalpy of reaction: $SmH_{2\pm x}+H_2 \Rightarrow SmH_{3-y}$	DTA	ΔH at given P	−	[47]
Hydrogen solubility in (α-Sm) at (α-Sm)/SmH_{2-x}	DTA	T at given x_H, P	+	[81]
boundary	equilibration analysis	x_H at given T, P	+	[81]
Hydrogen solubility in (α-Sm) at (α-Sm)/(β-Sm)	DTA	T at given x_H, P	+	[81]
boundary	equilibration analysis	x_H at given T, P	+	[81]
Hydrogen solubility in (β-Sm) at (β-Sm)/SmH_{2-x}	DTA	T at given x_H, P	+	[81]
boundary	equilibration analysis	x_H at given T, P	+	[81]
Hydrogen solubility in (β-Sm) at (α-Sm)/(β-Sm) boundary	DTA	T at given x_H, P	+	[81]
Hydrogen solubility in (β-Sm) at (β-Sm)/(γ-Sm) boundary	DTA	T at given x_H, P	+	[81]
Hydrogen solubility in (γ-Sm) at (β-Sm)/(γ-Sm) boundary	DTA	T at given x_H, P	+	[81]
(γ-Sm)–solidus	DTA	T at given x_H, P	+	[81]
Liquidus	DTA	T at given x_H, P	+	[81]
Hydrogen content in $SmH_{2\pm x}$ at (α-Sm)/SmH_{2-x}	XRD	x_H at given T, P	−	[84]
boundary	XRD	x_H at given T, P	−	[87]
Hydrogen content in $SmH_{2\pm x}$ at SmH_{2+x}/SmH_{3-y}	XRD	x_H at given T, P	−	[84]
boundary	TG	x_H at given T, P	−	[47]
	XRD	x_H at given T, P	−	[87]
	XRD	x_H at given T, P	−	[86]
	XRD	x_H at given T, P	+	[85]
Hydrogen content in SmH_{3-y} at SmH_{2+x}/SmH_{3-y}	XRD	x_H at given T, P	−	[84]
boundary	TG	x_H at given T, P	−	[47]
	XRD	x_H at given T, P	−	[87]
	XRD	x_H at given T, P	−	[86]
Hydrogen content in $SmH_{2\pm x}$ at $SmH_{2\pm x}$/H_2 boundary	TG	x_H at given T, P	+	[47]
Hydrogen content in SmH_{3-y} at SmH_{3-y}/H_2	TPA	x_H at given T, P	+	[91]
boundary	TG	x_H at given T, P	−	[47]
	XRD	x_H at given T, P	−	[87]
Equilibrium H_2 pressure, single (β-Sm) phase	P–V–T measurements	pH_2 at given x_H, T	+	[82]
Equilibrium H_2 pressure, (β-Sm)+SmH_{2-x}	P–V–T measurements	pH_2 at given T	+	[83]
		pH_2 at given T	+	[82]
Equilibrium H_2 pressure, single $SmH_{2\pm x}$ phase	P–V–T measurements	pH_2 at given x_H, T	+	[88]
		pH_2 at given x_H, T	+	[82]
		pH_2 at given x_H, T	−	[89]
		pH_2 at given x_H, T	+	[90]
Equilibrium H_2 pressure, SmH_{2+x}+SmH_{3-y}	P–V–T measurements	pH_2 at given T	+	[88]
		pH_2 at given T	+	[90]
		pH_2 at given T	−	[89]
Equilibrium H_2 pressure, single SmH_{3-y} phase	P–V–T measurements	pH_2 at given x_H, T	+	[88]
		pH_2 at given x_H, T	−	[90]

[a] This column indicates whether the values were used (+) or not used (−) in the optimization.

structure. The thermodynamic parameters for all phases were optimized simultaneously, but first $G^{\text{b.c.c.}}_{\text{Fe:H}}$ was represented as a linear function of temperature. Then, the coefficients c and d in Eq. (1) were successively taken into account. Note the splitting of the Gibbs energy expression for a hypothetical b.c.c. iron hydride into two temperature ranges (Table 1). This was done because the optimization did not take into account the behaviour of $G^{\text{b.c.c.}}_{\text{Fe:H}}$ well above the melting point of pure iron (1811 K). However, the phase diagram calculated in the whole temperature range (298–6000 K) for $P=10$ MPa showed that the b.c.c. phase becomes stable again above 2900 K. To avoid such artifact a linear extrapolation was performed between 1811 and 6000 K, while at the melting point of iron both functions have identical values.

5.5. The $Fe_{17}Sm_2$–H_2 system

The thermodynamic parameters for the interstitial solid solution $Fe_{17}Sm_2H_\delta$ were evaluated using equilibrium hydrogen pressure measurements [57]. It has been proven that the hydrogen solubility can be satisfactorily described within the sublattice model using one interaction parameter

Table 5
Selection of experimental information for the Fe–H system

Data description	Experimental method	T (K)	P (MPa)	Use[a]	Ref.
Hydrogen solubility in solid Fe, x_H at given T and P	P–V–T measurements	682–1409	0.1013	+	[92]
		1410–1811	0.1013	−	[92]
	P–V–T measurements	713–1432	0.1013	+	[93]
	P–V–T measurements	573–1473	0.1013	+	[94]
	P–V–T measurements	773–1595	0.1013	+	[95]
		1596–1811	0.1013	−	[95]
	GE[b] + VHE[c]	473–993	0.1013	+[f]	[96]
	GE + VHE	873–1100	0.1013	−	[53]
		1100–1644	0.1013	+	[53]
		663–763	2.2–13.8	+	[53]
		<663, >763	2.2–13.8	−	[53]
	P–V–T measurements	1473–1811	0.1013	+	[51]
	GE + VHE (SC)[d]	564–1423	0.1013	+[g]	[52]
	(PC)[e]	564–1423	0.1013	−	[52]
	GE + VHE	1250–1713	0.1013	+[h]	[71]
	GE + VHE	1373–1773	≤10	+	[54]
		873–1123	≤10	−	[54]
	GE + Volumetry	300–500	0.1013	−	[97]
	GE + VHE	293–773	0.1013	−	[98]
	GE + VHE	773–1073	0.1013	−	[99]
Hydrogen solubility in liquid Fe, x_H at given T and P	P–V–T measurements	1811–1923	0.1013	+	[92]
	P–V–T measurements	1811–1923	0.1013	−	[95]
	P–V–T measurements	1873	0.1013	+	[100]
	GE + VHE	1853–1943	0.1013	+	[101]
	P–V–T measurements	1811–1973	0.1013	+	[51]

[a] This column indicates whether the values were used (+) or not used (−) in the optimization.
[b] Gas equilibration.
[c] Vacuum hot extraction.
[d] Single-crystal.
[e] Polycrystal.
[f] The experimental errors are not given in the source—estimated $\Delta T = \pm 10$ K, $\Delta x_H = 5\%$.
[g] The experimental errors are not given in the source—estimated $\Delta T = \pm 5$ K, $\Delta x_H = 10\%$.
[h] The experimental errors are not given in the source—estimated $\Delta T = \pm 5$ K, $\Delta x_H = 20\%$.

(Eq. (5)), which is a linear function of temperature. Because the magnetic properties were measured only for end-members of solution ($Fe_{17}Sm_2$ and $Fe_{17}Sm_2H_5$) no magnetic interactions were taken into account assuming a linear variation of T_C and β with composition.

To complete the thermodynamic description, the Gibbs energy of formation for two fictitious compounds FeH with the rhombohedral α-Sm-type and h.c.p. structure as well as for the f.c.c. samarium monohydride were estimated (Table 1), since they enter into Eq. (3), but cannot be taken from assessments of Fe–H or H–Sm systems. In all of these phases hydrogen would occupy the interstitial positions; the interaction parameters between hydrogens and vacancies were set equal to those, which have been evaluated for (α-Sm), (β-Sm), and f.c.c. Fe, respectively. The interaction parameters between Fe and Sm atoms in the metallic sublattice of liquid, b.c.c., f.c.c., h.c.p., and (α-Sm) phase were adopted from Fe–Sm assessment (see above) independent of whether the interstitial sublattice is filled with hydrogens.

6. Results and discussion

6.1. The Fe–Sm system

Fig. 1 shows the calculated T–x phase diagram Fe–Sm together with the experimental points. A very good agreement is achieved. Compared to the previous work [10] the present thermodynamic description reproduces better the measured temperatures of the invariant reactions. The calculated liquidus curve shows noticeable deviations from the experimental points in the Fe-rich region. However, the precise determination of liquidus temperatures in the systems with rare earths is difficult. Apart from the evaporation of Sm and oxidation, the reaction between the molten alloy and a crucible (typically Al_2O_3) is unavoidable. At the same time, the invariant reaction isotherms can be ascertained with higher accuracy since they are usually detected as sharp peaks on DTA traces in a wide composition range.

All invariant equilibria in the Fe–Sm system are listed

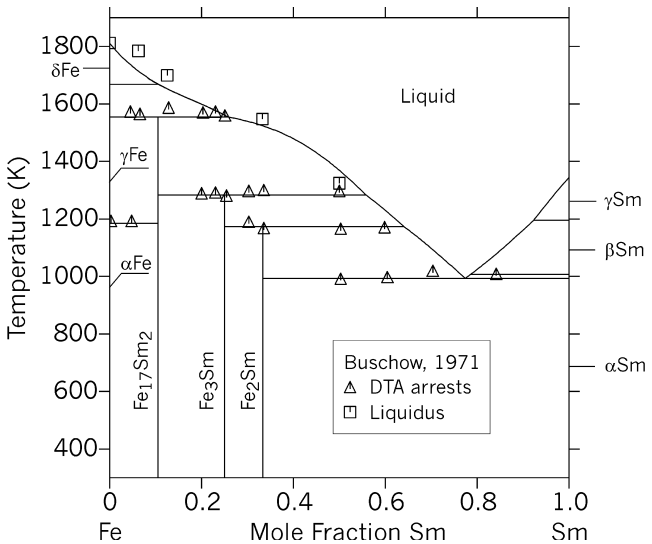

Fig. 1. Calculated Fe–Sm phase diagram with the experimental measurements superimposed.

in Table 6. The calculated and measured thermochemical properties of $Fe_{17}Sm_2$ are compared in Table 7. One can see that the present thermodynamic description is consistent with the standard entropy value determined by integration of the experimental heat capacity data assuming the same relative standard deviation (5%) for $°S_{298}$ as for C_P (the actual uncertainty for the entropy may be even higher due to the accumulation of errors during integration). The calculated heat capacity itself is beneath the measured values, but still within the confidence limits. Fig. 2 illustrates graphically the specific heat of $Fe_{17}Sm_2$ in the range from 0 to 700 K. The peak is predicted at the critical temperature of magnetic ordering. Note that the magnetic contribution to the heat capacity (and to the Gibbs energy) becomes significant already at the temperatures, which are

Table 7
Thermochemical properties of $Fe_{17}Sm_2$ measured and calculated in the present work

Property	T (K)	Measured	Calculated
Entropy (J mol^{-1} K^{-1})[a]	298.15	36.6±1.5	35.18
Heat capacity (J mol^{-1} K^{-1})[b]	272	29.1	25.95
	277	29.4	26.42
	282	29.7	26.92
	287	29.9	27.45
	293	30.2	28.00
	298	30.3	28.57

[a] Integrated heat capacity data.
[b] The measurement accuracy is <5%.

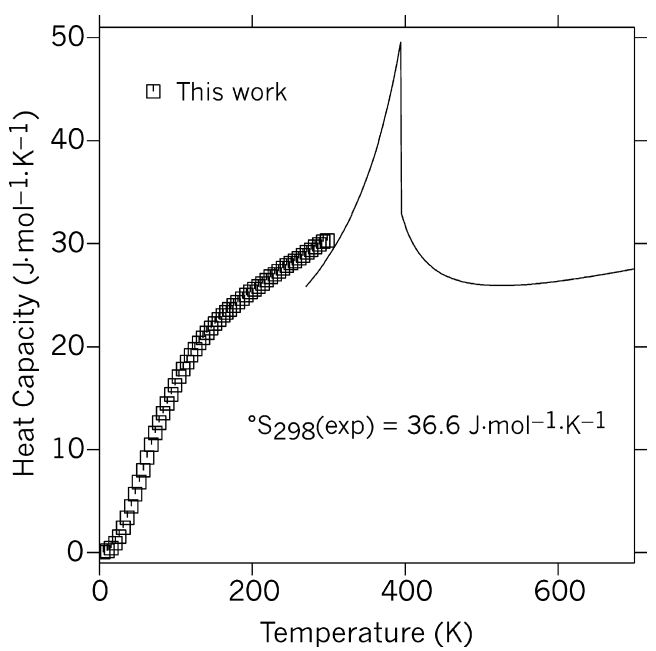

Fig. 2. Measured and calculated heat capacity of $Fe_{17}Sm_2$.

Table 6
Invariant reactions in the Fe–Sm system

Reaction	Mole fractions Sm in phases			T (K)	Ref.
Peritectic:	–	–	–	1553±10	[12]
Liquid + f.c.c. Fe ⇒ $Fe_{17}Sm_2$	0.247	0	0.105	1558	This work
Peritectic:	–	–	–	1283±10	[12]
Liquid + $Fe_{17}Sm_2$ ⇒ Fe_3Sm	0.544	0.105	0.250	1287	This work
Peritectic:	–	–	–	1173±10	[12]
Liquid + Fe_3Sm ⇒ Fe_2Sm	0.631	0.250	0.333	1172	This work
Eutectic:	0.75	–	–	993±10	[12]
Liquid ⇒ Fe_2Sm + α-Sm	0.768	0.333	1.000	997	This work

100 K below T_C. If one presumes that the C_P measurements were influenced by systematic errors and the true heat capacity is about 95% of the experimentally determined one, the calculated curve will be a reasonable extension of the low-temperature data.

The thermodynamic description of the Fe–Sm system can be further refined with more experimental information on the thermochemical properties, such as enthalpy and heat capacity of the intermetallic compounds in the extended temperature range or the heat of mixing in the liquid state.

6.2. The H–Sm system

The calculated H–Sm phase diagram for the hydrogen pressure of 101 325 Pa is compared in Fig. 3a with experimental measurements. Almost all phase diagram data were reproduced by calculations within the experimental uncertainties, whether or not they were used to evaluate the thermodynamic parameters. A few points [84,87], which do not match the calculated border of homogeneity range for samarium dihydride phase in equilibrium with (α-Sm) cannot be reconciled with the corresponding equilibrium hydrogen pressures and, therefore, they were omitted in the optimization (see Table 4). It should be noted, however, that the calculated limit of hydrogen solubility in the liquid, (β-Sm), and (γ-Sm) is based on the last measured point at 28 at% H in the work of Beadry and Gschneidner [81] and does not necessarily represent the maximum achievable concentration. On the other hand, for the (β-Sm) phase the maximum hydrogen solubility does not exceed 40 at% [83]. The invariant reactions in H–Sm system for a hydrogen pressure of 101 325 Pa are given in Table 8. The calculated values show a very good agreement with the experimentally determined temperature and composition of phases at the eutectoid and gas–peritectoid equilibrium. At the same time, the enthalpy of the gas–peritectoid reaction $SmH_{2.40} + 0.225H_2 = SmH_{2.85}$ [47,48] was not successfully reproduced by calculations (Table 8). The most probable reason of such disagreement is that the composition limits of the hexagonal to cubic transition calculated in the present work differ from those given in Ref. [48].

The present assessment confirmed the existence of a peritectic and a peritectoid equilibrium, in which the (γ-Sm) and (β-Sm) phases are formed, respectively, as suggested by Manchester and Pitre [46], who used a question mark instead of SmH_{2-x}. Furthermore, the calculations predict a new gas–peritectic reaction: liquid + $H_2 \Rightarrow SmH_{2\pm x}$. Thus, the melting of samarium dihydride should be incongruent and result in evolving a relatively large amount of the hydrogen gas, while the Sm-rich liquid remains as a condensed phase. To check the validity of predictions further DTA/TG experiments between 30 and 80 at% H are to be performed.

The boundary of a homogeneity region for $SmH_{2\pm x}$ phase is very steep in hypostoichiometric region (i.e. in equilibrium with interstitial solid solution phases and liquid), the hydrogen content slightly decreasing with increase of temperature. This is apparently related to the thermal disordering, so that a fraction of vacant tetrahedral sites increases. At the same time, the hydrogen content in SmH_{2+x}, which is equilibrated with the gas phase decreases more rapidly with increasing temperature. This indicates a weaker bonding between Sm and H in octahed-

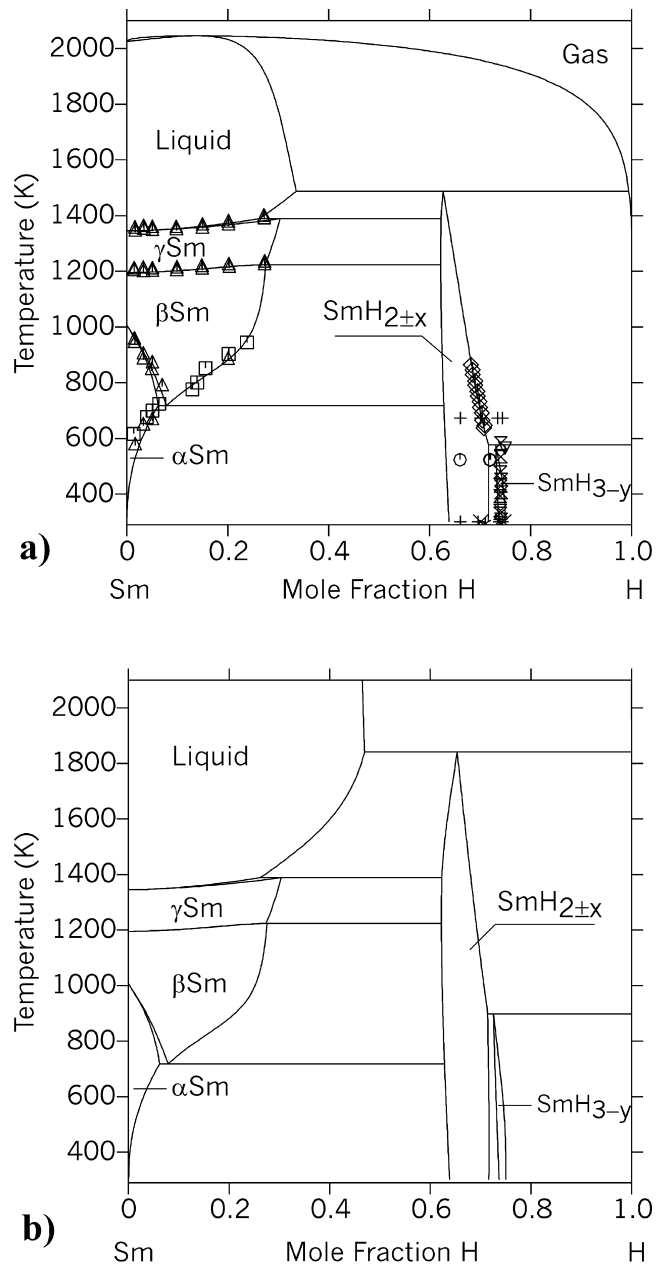

Fig. 3. Calculated H–Sm phase diagram in comparison with experimental data: (a) $P = 101\,325$ Pa; (b) $P = 10$ MPa. Beadry and Gschneidner [81]: (△) DTA, (□) Solubility; Mintz et al. [47]: (◇) TG SmH_2–Gas, (⧖) TG SmH_3–Gas; (▽) Rodin and Grushina [91]; (+) Greis et al. [87]; (×) Zogal et al. [86]; (✱) Daou et al. [85]; (⏲) Pebler and Wallace [84].

Table 8
Invariant reactions in the H–Sm system at 101 325 Pa hydrogen pressure

Reaction	Mole fractions H in phases			T (K)	Ref.
Eutectoid:	0.100	0.080	–	703±20	[81]
(β-Sm)⇒(α-Sm)+SmH_{2-x}	0.079	0.062	0.628	718	This work
Gas–peritectoid:	0.706	1.000	0.740	587±20	[47,48]
$SmH_{2\pm x}+H_2 \Rightarrow SmH_{3-y}$	±0.005		±0.005		
($\Delta^\circ H_{tr}=-16.7\pm1.7$ kJ mol^{-1})					
$\Delta^\circ H_{tr}=+7.2$ kJ mol^{-1}	0.716	1.000	0.731	580	This work
Peritectoid:					
(γ-Sm)+SmH_{2-x}⇒(β-Sm)	0.274	0.622	0.275	1224	This work
Peritectic:					
Liquid+SmH_{2-x}⇒(γ-Sm)	0.262	0.623	0.304	1389	This work
Gas–peritectic:					
Liquid+$H_2 \Rightarrow SmH_{2-x}$	0.340	0.994	0.627	1493	This work

ral sites, which become completely vacant at the temperatures close to the point of peritectic decomposition of SmH_{2-x}. Note, that the hydrogen concentration in samarium dihydride in equilibrium with liquid and gas is lower than the stoichiometric one.

The predicted phase diagram at 10 MPa pressure (Fig. 3b) show the significant increase of the temperatures, up to which the samarium hydrides remain stable. Unfortunately, there are no measurements for comparison.

The calculated equilibrium hydrogen pressures in H–Sm system match very well to the experimental data in the phase regions from pure samarium up to samarium trihydride, except those of Ohki et al. [89], which were not used in the optimization (Fig. 4a–d). The temperatures are not indicated on the isotherms to provide more clarity, since it is straightforward to find out the correspondence. The small plateau in Fig. 4a, which occurs in the Sm-rich region is due to the coexistence of (α-Sm) and (β-Sm). The increase of the calculated equilibrium hydrogen pressure upon filling the interstitials in samarium dihydride is not so steep as found experimentally in Ref. [82] (Fig. 4b), but the agreement is still satisfactory taking the accuracy of P–V–T measurements into account. The calculations predict that the trihydride phase approaches the stoichiometric composition only at very high pressures or at very low temperatures (Fig. 4d). The discrepancy with experiments is not, however, appreciably larger, than the experimental errors in determination of hydrogen content. The present thermodynamic description of the H–Sm system is in best agreement with the data of Messer and Park [88].

6.3. The Fe–H system

The calculated phase diagram of the Fe–H system at the hydrogen pressure of 101 325 Pa is shown in Fig. 5a. The experimental points are not included in this figure, since they would overlap each other and completely mask the calculated lines. The dissolution of hydrogen at the normal pressure does not alter the temperatures of the allotropic transformations of iron within ±1 K.

The solubility of hydrogen in iron at 101 325 Pa pressure is presented in Fig. 5b–d to show all available literature data. Excellent agreement with the experiments selected for the optimization can be seen in these figures. In the low-temperature region (300–700 K, Fig. 5b), the calculated hydrogen solubility in b.c.c. iron is consistent with most of the measurements [52,71,92,96]. At the same time, there are two literature sources [97,98], where the noticeably higher solubility has been reported. Following Kiuchi and McLellan [50] it might be concluded that these experiments were influenced by surface-related effects.

Fig. 5c shows the hydrogen solubility in the medium temperature range (700–1300 K). Again, most of the experimental data are well reproduced by calculations. The best agreement exists between 700 and 1000 K, while the scattering of experimental points becomes generally larger as the temperature increases. A small inflection can be seen on the calculated solubility curve at the Curie temperature of b.c.c. iron. This could not be, of course, observed experimentally because the corresponding change is as large as the uncertainty of the measurements. In the f.c.c. iron the hydrogen solubility abruptly increases by about 60% and then grows more rapidly with temperature. Although the equilibrium hydrogen content in f.c.c. Fe proposed by Da Silva and McLellan [71] is much smaller then calculated, the latter matches excellently all other experimental data (Fig. 5c). The data of Sieverts et al. [95] exhibit a jump already at 1160–1170 K indicating probably a systematic error in their measurements of temperature. Indeed, the agreement between data in Ref. [95] and calculated curve becomes appreciably better, if the corresponding datapoints will be shifted along the horizontal axis by +20 K.

The high-temperature part of the solubility curve is

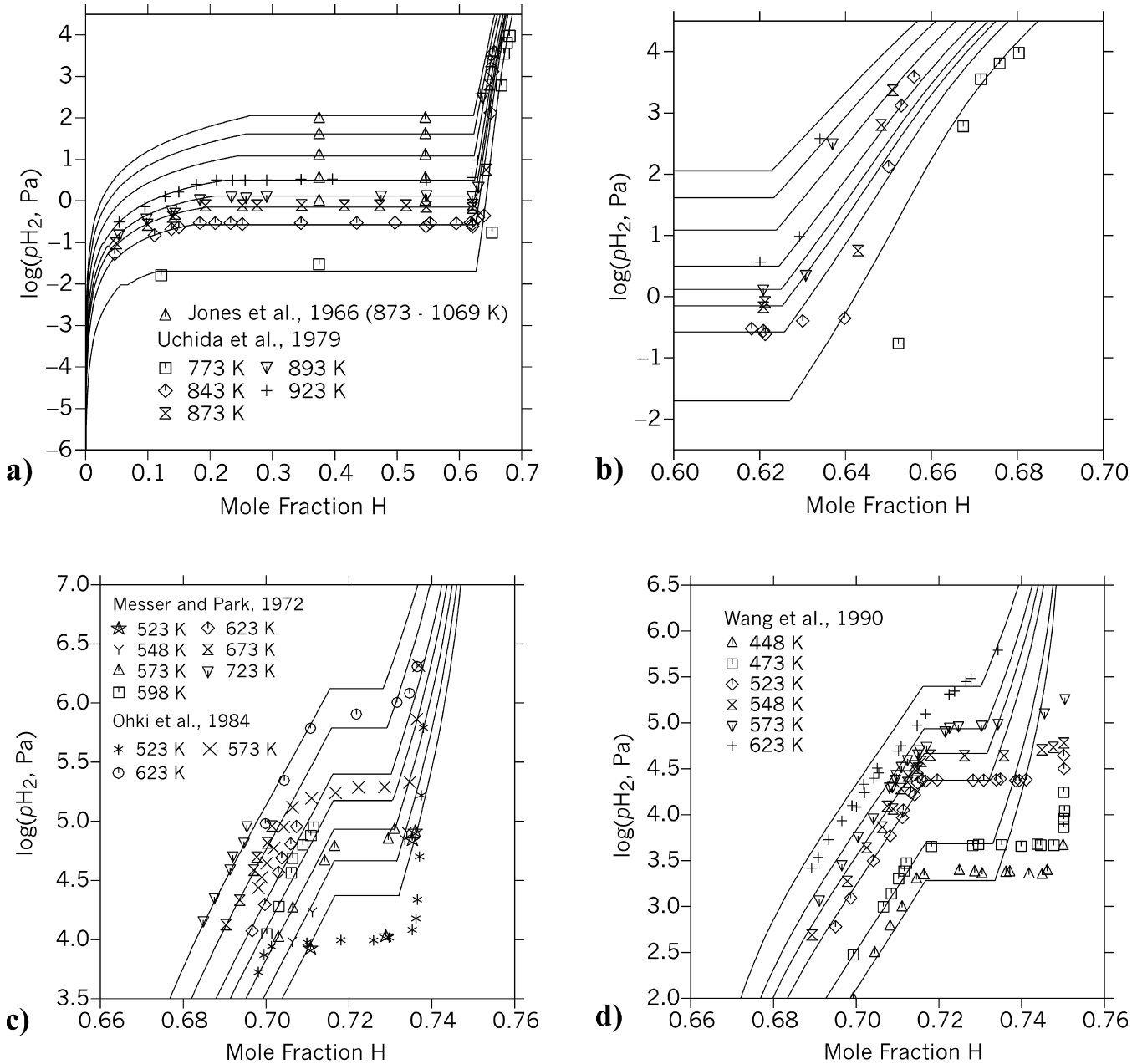

Fig. 4. Calculated equilibrium hydrogen pressures in H–Sm system for the region between Sm metal and SmH_{2-x} (a, b) and between $SmH_{2\pm x}$ and SmH_{3-y} (c, d) compared with experimental measurements.

presented in Fig. 5d. One can see the small drop of the hydrogen content due to the γ⇔δ transition that has been overlooked in some experimental works and a large (about two times) increase of solubility at the melting point of iron, which has been observed in all studies extending above 1811 K. The present calculations show that decreasing solubility in δ-Fe is an intrinsic thermodynamic property of the iron–hydrogen system. The agreement between experimental and calculated solubility of hydrogen in solid iron is quite satisfactory excluding the data of Da Silva and McLellan [71] and applying the temperature correction to the measurements of Sieverts et al. [95] (see above). For the liquid iron, the calculations match perfectly the measurements of Schenk and Lange [51], which lie in the middle between other data scattering by about ±30%. However, no preference can be given for the particular data set without more accurate measurements of the hydrogen solubility in the liquid iron.

Fig. 6 shows the Fe–H phase diagram calculated at a hydrogen pressure of 10 MPa. The temperature of the α⇔γ transition decreases and of γ⇔δ transition increases by 9 and 15 K, respectively. The melting point of iron is

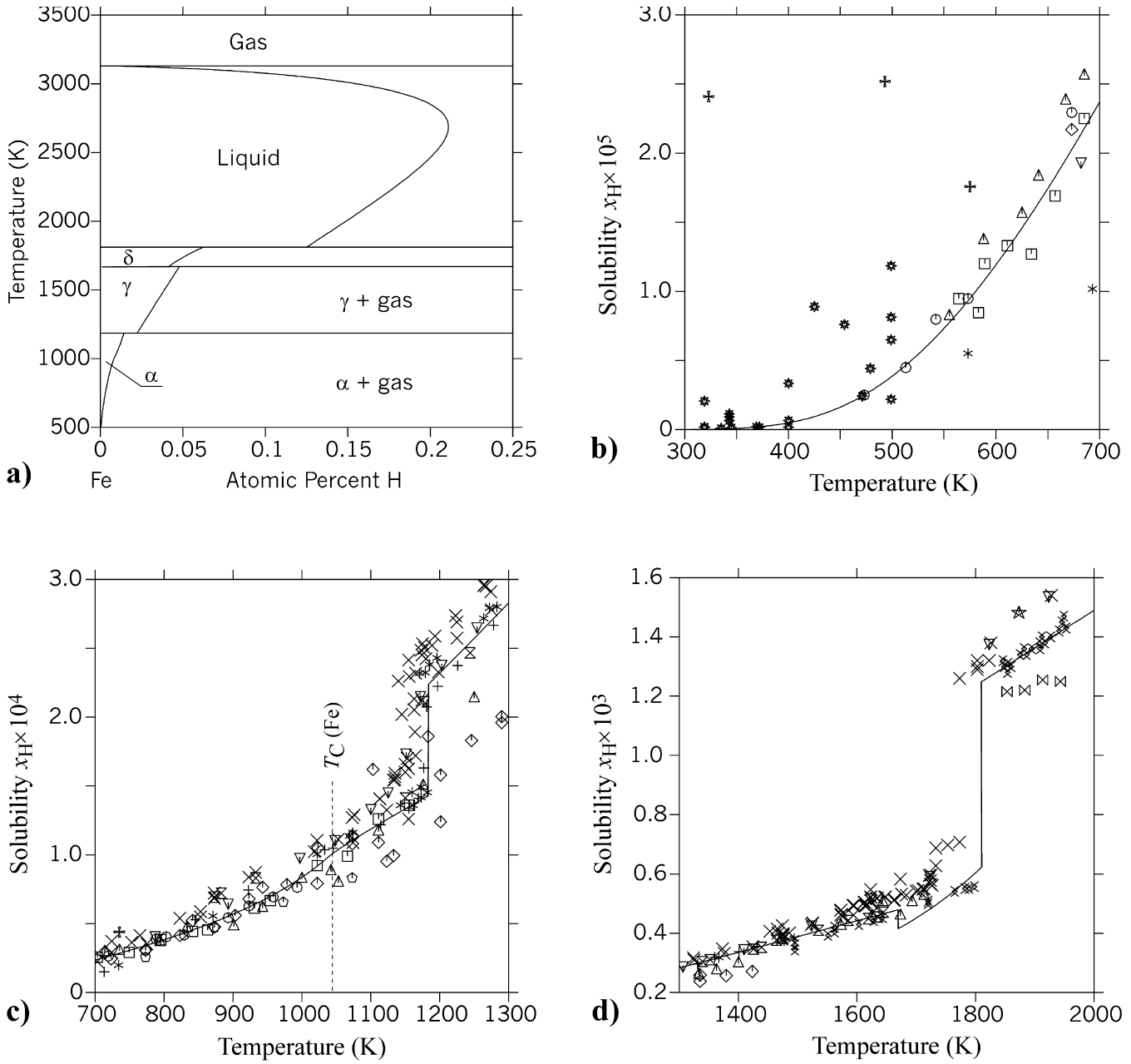

Fig. 5. Calculated phase diagram of the Fe–H system (a) and the solubility of hydrogen in iron (b–d) at 101 325 Pa pressure with the experimental measurements superimposed: (▽) Sieverts [92]; (+) Martin [93]; (✳) Luckmeyer-Hasse and Schenck [94]; (×) Sieverts et al. [95]; (⊙) Eichenauer et al. [96]; (☆) Busch and Dodd [100]; (⧖) Hill and Johnson [53]; (⋈) Schenk and Wünsch [101]; (✳) Schenck and Lange [51]; (△) Da Silva and McLellan [71] (single-crystal); (□) Da Silva et al. [52] (single-crystal); (◇) Da Silva et al. [52] (polycrystal); (✱) Louthan et al. [97]; (✢) Choo et al. [98]; (⬠) Lee et al. [99].

lowered by 12 K. This is in qualitative agreement with the observations of Shapovalov et al. [54].

For the pressure dependence of the hydrogen solubility at constant temperatures, calculations are compared with the measurements of Shapovalov et al. [54] in Fig. 7a and with those of Hill and Johnson [53] in Fig. 7b. Good agreement with data in Ref. [54] exists between 1373 and 1773 K up to 10 MPa as long as the experimental solubility obeys Sieverts law. It is believed that at lower temperatures the measurements were influenced by systematic errors since the solubility is not consistent with the corresponding values at 101 325 Pa pressure. The data of Hill and Johnson [53] were reproduced by calculations within the limits of experimental accuracy only for 663 and 763 K (Fig. 7b), while at higher and lower temperatures much greater standard deviations should be assumed.

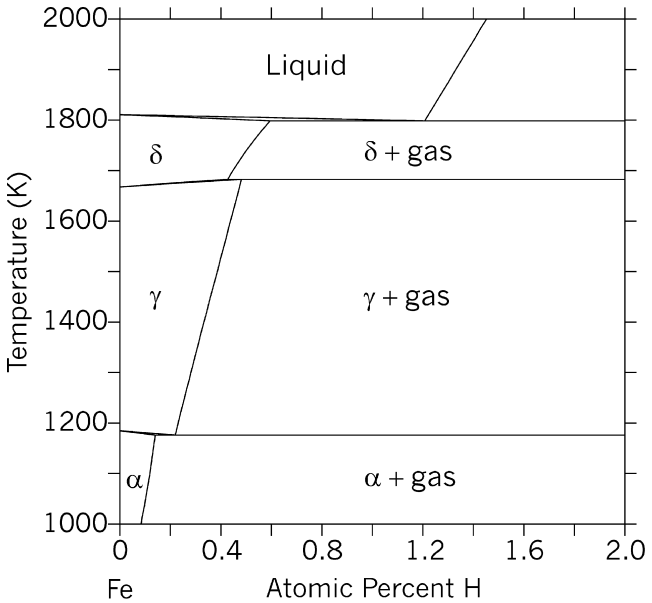

Fig. 6. Calculated phase diagram of the Fe–H system at 10 MPa hydrogen pressure.

As indicated by the authors [53], the low-temperature data are less confident because of the presence of residual hydrogen, which becomes trapped by impurities.

6.4. The $Fe_{17}Sm_2$–H_2 system

$Fe_{17}Sm_2H_\delta$ as well as $Fe_{17}Sm_2$ is thermodynamically unstable against disproportionation in hydrogen atmosphere up to the temperature of recombination. The lines of limited hydrogen solubility in $Fe_{17}Sm_2$ were calculated for different pressures (Fig. 8) taking only $Fe_{17}Sm_2H_\delta$ and the vapor phase into account, while other phases were excluded from consideration. Such method is often used to calculate the metastable phase diagrams. The kink on the solubility curves is predicted at the Curie temperature, which varies linearly with composition. The only experimentally determined hydrogen solubility [60] differs from the calculated value by as much as one hydrogen atom per formula unit. In the present work, the data of Koeninger et al. [57] was given preference because they represent equilibrium isotherms.

Fig. 9 represents the calculated hydrogen partial pressures in the system $Fe_{17}Sm_2H_\delta$–H_2 superimposed with the experimental points. An excellent agreement exists in the region $T \geq 573$ K. At lower temperatures, the only high-pressure range is well reproduced by calculations. However, the authors themselves [57] pointed out that experimental difficulties exist to achieve the equilibrium below 573 K and could not fit their own data in this region using a model of an ideal interstitial solution. Therefore, it might be concluded that the hydrogen solubility in $Fe_{17}Sm_2$ is well described by the present calculations.

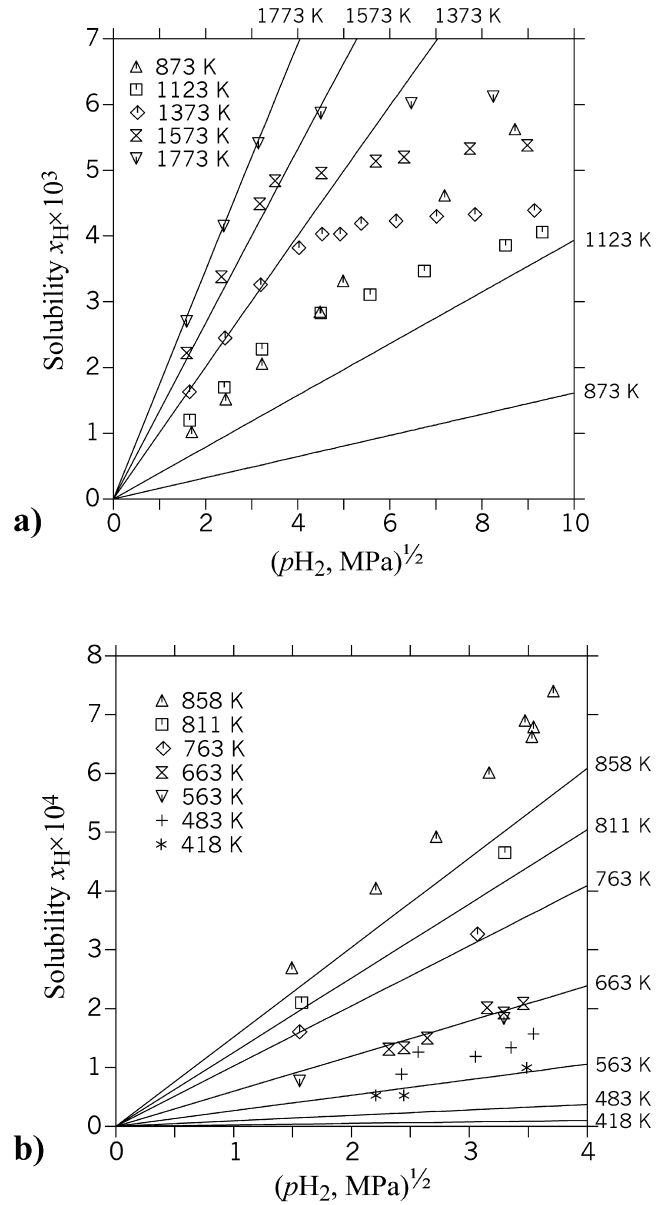

Fig. 7. Calculated pressure dependence of the hydrogen solubility in iron at constant temperatures compared with the measurements of Shapovalov et al. [54] (a) and of Hill and Johnson [53] (b).

6.5. Thermodynamic modeling of the HDDR process

With the thermodynamic parameters presented in Table 1, the thermodynamics of the HDDR process can be described. In Fig. 10 the equilibrium T–P phase diagram $Fe_{17}Sm_2H_\delta$–H_2 is shown. A good matching to a few experimental points, which represent the equilibrium recombination temperature as a function of hydrogen pressure was obtained (Table 9). At hydrogen pressures higher than about 435 kPa the recombination is not possible, since it would require a temperature, at which the $Fe_{17}Sm_2$ phase is not stable against decomposition into f.c.c. Fe and liquid. In accordance with the thermodynamics of H–Sm

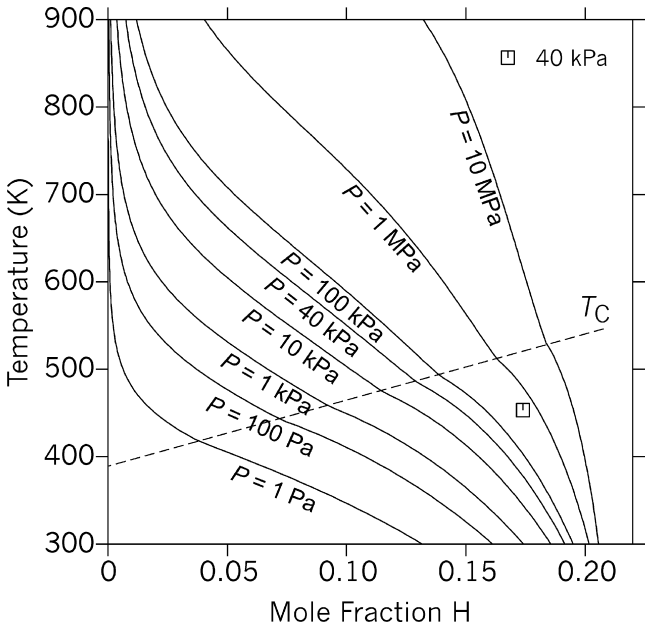

Fig. 8. Calculated hydrogen solubility in $Fe_{17}Sm_2$ at different pressures in comparison with the measurement of Ref. [60].

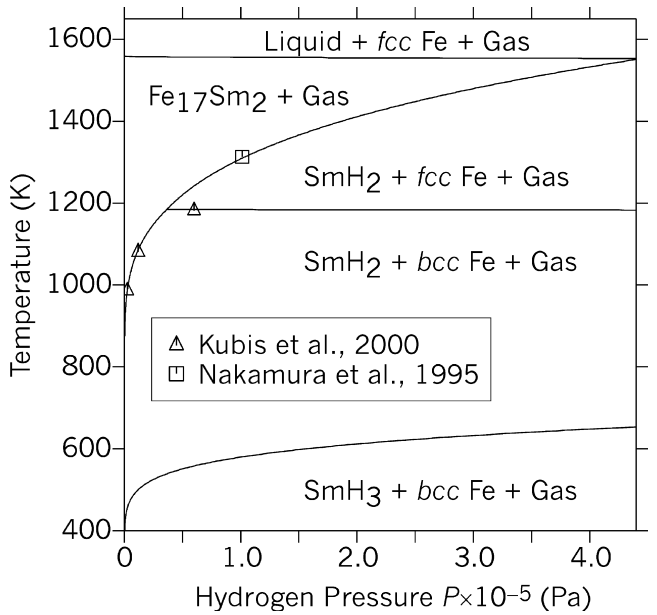

Fig. 10. Equilibrium T–P phase diagram of the $Fe_{17}Sm_2H_\delta$–H_2 system compared with experimental data.

system, the calculations predict that at low temperatures the stable phases in equilibrium with hydrogen gas become SmH_{3-y} and b.c.c. Fe. Experiments performed by Christodoulou and Takeshita [19] can be considered as experimental evidence for the phase transition $SmH_{2+x}+H_2 \Rightarrow SmH_{3-y}$: they observed a jump at ~473 K on the absorption curve obtained during cooling of $Fe_{17}Sm_2$, which was previously disproportionated at 1223 K in hydrogen atmosphere.

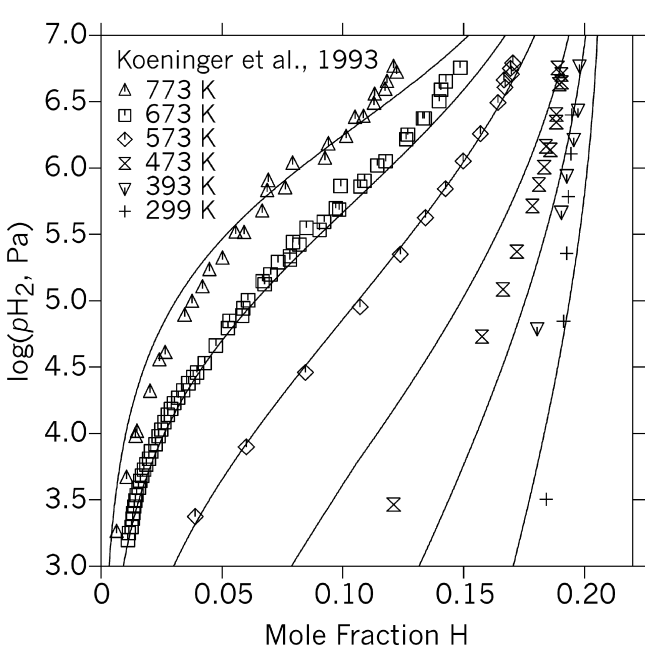

Fig. 9. Calculated equilibrium hydrogen partial pressures in the system $Fe_{17}Sm_2H_\delta$–H_2 superimposed with the experimental points.

The temperature of the disproportionation reaction $Fe_{17}Sm_2H_\delta + H_2 \Rightarrow SmH_{2\pm x} + Fe$ cannot be calculated using equilibrium thermodynamics, since the existence of $Fe_{17}Sm_2H_\delta$ at low temperatures is only because of the activation barrier (probably due to the nucleation of samarium hydrides). To obtain evidence on how the disproportionation temperature will change with increasing pressure, the driving force for the disproportionation reaction was calculated as a function of temperature and hydrogen pressure (Fig. 11). One can see that at high temperatures the driving force becomes more negative with increasing pressure, so that the disproportionation temperature should decrease, as it was experimentally observed [58]. At the same time, the curves are crossing each other upon decreasing temperature and the behaviour is completely opposite around 700 K. This effect becomes more pronounced with increasing hydrogen pressure. Such behaviour is obviously due to the increasing stability of $Fe_{17}Sm_2H_\delta$ as a result of a drastic increase in δ at low temperatures and high pressures (Fig. 8). Thus, the calcu-

Table 9
Experimental and calculated temperatures (K) of the invariant recombination reaction: $Fe + SmH_{2\pm x} \Rightarrow Fe_{17}Sm_2 + H_2$

P (kPa)	T (measured)	T (calculated)	Ref.
2.5	986	985	[63]
12	1081	1086	[63]
60	1182	1242	[63]
101.325	1313	1310	[61]

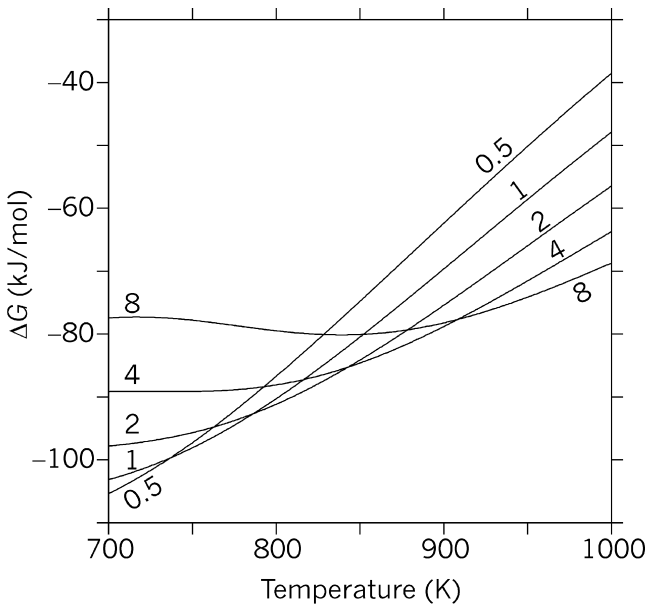

Fig. 11. The driving force for the disproportionation reaction: $Fe_{17}Sm_2H_\delta + 0.5 \cdot (2n + 17m - \delta)H_2 = 2SmH_n + 17FeH_m$ calculated for different hydrogen pressures (numbers on the curves $\times 10^{-5}$, Pa).

lations predict that the disproportionation temperature will increase with increasing pressure, if the latter is higher than some threshold level. This has to be still verified experimentally by DTA measurements at higher pressures.

7. Conclusions

- Heat capacity of the intermetallic compound $Fe_{17}Sm_2$ has been measured between 5 and 298 K and the standard entropy has been calculated.
- The Fe–Sm system has been reassessed in order to obtain the thermodynamic description, which is consistent with the commonly accepted unary data and to take the new measurements of this study into account. The obtained thermodynamic parameters reproduce experimental data very well.
- Two binary systems Fe–H and H–Sm have been thermodynamically assessed for the first time. In both cases, a self-consistent thermodynamic description for pressures up to 10 MPa was obtained.
- The thermodynamics of hydrogen in the interstitial solid solution phase $Fe_{17}Sm_2H_\delta$ can be successfully described with the use of compound energy formalism. The hydrogen content (δ) increases drastically at low temperatures and high pressures up to the maximum value $\delta = 5$.
- Combining the obtained Gibbs energy functions for the individual phases, the equilibrium temperature–pressure phase diagram of the $Fe_{17}Sm_2$–H_2 system has been calculated. The calculated temperature of the recombination reaction: $Fe + SmH_{2\pm x} \Rightarrow Fe_{17}Sm_2 + H_2$, which is an important part of the HDDR process is in good agreement with the experimental measurements. At hydrogen pressures higher than about 435 kPa the recombination is not possible, since it would require the temperature, at which the $Fe_{17}Sm_2$ phase decomposes into f.c.c. Fe and liquid. The calculations predict that the disproportionation temperature will increase with increasing pressure, if the latter is higher than some threshold level.

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